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		$$ax^2 + bx + c = 0$$ are given by the following equations : $$x_1 = \frac{-b + \sqrt[2]{(b^2 - 4ac)}}{2a}$$ $$x_2 = \frac{-b - \sqrt[2]{(b^2 - 4ac)}}{2a}$$ Write a function to calculate the roots. The function must use two pointer parameters, one to receive the coefficients a, b, and c, and the other to send the roots to the calling function.		
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Structure And Union

1. Define a structure data type called **times_struct** containing three members integer **hour**, integer **minute** and integer **second**. Develop a program that of would assign values to the individual members and display the time in the following form:

   16:40:51

```
#include<stdio.h>

struct time_struct
{
   int hour, minute, second;
}ts;

int main()
{
   ts.hour = 16;
   ts.minute = 40;
   ts.second = 51;

   printf("%02d : %02d : %02d \n", ts.hour, ts.minute, ts.second);
   return 0;
}
```

Output :-

```
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Structure And Union$ gcc problem1.c -o problem1
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Structure And Union$ ./problem1
16 : 40 : 51
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Structure And Union$
```

2. Modify the above program such that a function is used to input values to the members and another function to display the type time.

```
#include <stdio.h>

struct time_struct
{
   int hour, minute, second;
}ts;

void input_time(struct time_struct * ts)
{
   printf("Enter hour : ");
   scanf("%d", &ts->hour);
   printf("Enter Minute : ");
   scanf("%d", &ts->minute);
   printf("Enter Second : ");
   scanf("%d", &ts->second);
}

int main()
{
   input_time(&ts);
   printf("%02d : %02d : %02d \n", ts.hour, ts.minute, ts.second);
   return 0;
}
```

Output :-

```
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Structure And Union$ gcc problem2.c -o problem2
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Structure And Union$ ./problem2
Enter hour : 16
Enter Minute : 40
Enter Second : 51
16 : 40 : 51
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Structure And Union$
```

3. Design a function **update** that would accept the data structure designed in Program 1 and increments time by one second and returns the new time.

```
#include <stdio.h>

struct time_struct
{
   int hour, minute, second;
}ts;

void input_time(struct time_struct *ts)
{
   printf("Enter hour : ");
   scanf("%d", &ts->hour);
   printf("Enter Minute : ");
   scanf("%d", &ts->minute);
   printf("Enter Second : ");
   scanf("%d", &ts->second);
}

int update(struct time_struct *ts)
{
   ts->second++;
   if (ts->second == 60)
   {
      ts->second = 00;
      ts->minute++;

      if (ts->minute == 60)
      {
         ts->minute = 00;
         ts->hour++;

         if( ts->hour == 24)
            ts-> hour = 0;
      }
   }
}

int main()
{
   input_time(&ts);
```

```
    printf("Before Updation : \n%d : %d : %d \n", ts.hour, ts.minute, ts.second);
    update(&ts);
    printf("After Updation (+1s): \n%d : %d : %d \n", ts.hour, ts.minute, ts.second);
    return 0;
}
```

Output :-

```
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Structure And Union$ gcc problem3.c -o problem3
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Structure And Union$ ./problem3
Enter hour : 16
Enter Minute : 40
Enter Second : 51
Before Updation :
16 : 40 : 51
After Updation (+1s):
16 : 40 : 52
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Structure And Union$
```

4. Define a structure data type named **date** containing three integer members **day**, **month** and **year**. Develop an interactive modular program to perform the following tasks:
    a. To read data into structure members by a function
    b. To validate the date entered by another function
    c. To print the date in the format

    April 29, 2002

    By a third function.
    The input data should be three integers like 29, 4, and 2002 corresponding to day, month, and year.

```
#include <stdio.h>
#include <string.h>

struct date
{
   int day;
   int month;
   int year;
}d;

void input_date(struct date *d)
{
   printf("Enter Day (1-31): ");
   scanf("%d", &d->day);
   printf("Enter Month (1-12): ");
   scanf("%d", &d->month);
   printf("Enter Year (e.g., 2002): ");
   scanf("%d", &d->year);
}

int leap_year(int year)
{
   if ((year % 4 == 0 && year % 100 != 0) || (year % 400 == 0))
      return 1;
   return 0;
}

int check_date(struct date d)
{
   int days[] = {0, 31, 28, 31, 30, 31, 30, 31, 31, 30, 31, 30, 31};
```

```
    if (d.year < 1 || d.month < 1 || d.month > 12 || d.day < 1 || d.day > 31)
        return 0;

    if (leap_year(d.year))
        days[2] = 29;

    if (d.day > days[d.month])
        return 0;

    return 1;
}

void print_date(struct date d)
{
    char month_names[13][15] = {"", "January", "February", "March", "April", "May", "June",
"July", "August", "September", "October", "November", "December"};

    printf("Formatted Date: %s %d, %d\n", month_names[d.month], d.day, d.year);
}

int main()
{
    int validation = 0;
    input_date(&d);

    if (check_date(d))
    {
        printf("\nSuccess! The date entered is a valid date.\n");
        print_date(d);
    }
    else
        printf("\nError! The date entered is a invalid date.\n");

    return 0;
}
```

Output :-

```
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Structure And Union$ gcc problem4.c -o problem4
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Structure And Union$ ./problem4
Enter Day (1-31): 24
Enter Month (1-12): 2
Enter Year (e.g., 2002): 2026

Success! The date entered is a valid date.
Formatted Date: February 24, 2026
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Structure And Union$
```

5. Design a function **update** that accepts the **date** structure designed in Program 4 to increment the date by one day and return the new date. The following rules are applicable:
    a. if the date is the last day in month , month should be incremented
    b. if it is the last day in December, the year should be incremented
    c. There are 29 days in February of a leap year

```
#include <stdio.h>

struct date
{
   int day;
   int month;
   int year;
} d;

void input_date(struct date *d)
{
   printf("Enter Day (1-31): ");
   scanf("%d", &d->day);
   printf("Enter Month (1-12): ");
   scanf("%d", &d->month);
   printf("Enter Year (e.g., 2002): ");
   scanf("%d", &d->year);
}

int leap_year(int year)
{
   if ((year % 4 == 0 && year % 100 != 0) || (year % 400 == 0))
      return 1;
   return 0;
}

int check_date(struct date d)
{
   int days[] = {0, 31, 28, 31, 30, 31, 30, 31, 31, 30, 31, 30, 31};

   if (d.year < 1 || d.month < 1 || d.month > 12 || d.day < 1 || d.day > 31)
      return 0;

   if (leap_year(d.year))
      days[2] = 29;
```

```
   if (d.day > days[d.month])
      return 0;

   return 1;
}

int update_date(struct date *d)
{
   int days[] = {0, 31, 28, 31, 30, 31, 30, 31, 31, 30, 31, 30, 31};

   if (leap_year(d->year))
      days[2] = 29;

   d->day++;
   if (d->day > days[d->month])
   {
      d->day = 1;
      d->month++;

      if (d->month > 12)
      {
         d->month = 1;
         d->year++;
      }
   }
   return 1;
}

void print_date(struct date d)
{
   char *month_names[] = {"", "January", "February", "March", "April", "May", "June",
"July", "August", "September", "October", "November", "December"};

   printf("%s %d, %d\n", month_names[d.month], d.day, d.year);
}

int main()
{
   while (1)
   {
      input_date(&d);

      if (!check_date(d))
```

```
        {
            printf("\nError! Invalid Date Entered\n");
            continue;
        }
        else
        {
            printf("\nSuccess! Valid Date Entered\n");
            break;
        }
    }

    update_date(&d);
    printf("Updated Date ( +1 day) : ");
    print_date(d);

    return 0;
}
```

Output :-

```
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Structure And Union$ gcc problem5.c -o problem5
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Structure And Union$ ./problem5
Enter Day (1-31): 15
Enter Month (1-12): 8
Enter Year (e.g., 2002): 1945
Updated Date ( +1 day) : August 16, 1945
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Structure And Union$
```

6. Modify the input function used in Program 4 such that it reads a value that represents the date in the form of a long integer , like 19450815 for the date 15 - 8 - 1945 (August 15 1945) and assigns suitable value to the members **day** , **month** , and **year**.

```
#include <stdio.h>

struct date
{
   int day, month, year;
} d;

void read_date(struct date *d)
{
   long int date;
   printf("Enter date in YYYYMMDD format: ");
   scanf("%ld", &date);

   d->year = date / 10000;
   d->month = (date % 10000) / 100;
   d->day = date % 100;
}

int leap_year(int year)
{
   if ((year % 4 == 0 && year % 100 != 0) || (year % 400 == 0))
      return 1;
   return 0;
}

int check_date(struct date d)
{
   int days[] = {0, 31, 28, 31, 30, 31, 30, 31, 31, 30, 31, 30, 31};

   if (d.year < 1 || d.month < 1 || d.month > 12 || d.day < 1 || d.day > 31)
      return 0;

   if (leap_year(d.year))
      days[2] = 29;

   if (d.day > days[d.month])
      return 0;
```

```
    return 1;
}

void print_date(struct date d)
{
    printf("\nYear = %04d \nMonth = %02d \nDay = %02d\n", d.year, d.month, d.day);
}

int main()
{
    while (1)
    {
        read_date(&d);

        if (!check_date(d))
        {
            printf("\nError! Invalid Date Entered\n");
            continue;
        }
        else
        {
            print_date(d);
            break;
        }
    }
    return 0;
}
```

Output :-

```
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Structure And Union$ gcc problem6.c -o problem6
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Structure And Union$ ./problem6
Enter date in YYYYMMDD format: 1045815

Error! Invalid Date Entered
Enter date in YYYYMMDD format: 10450815

Year = 1045
Month = 08
Day = 15
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Structure And Union$
```

7. Add a function called **nextdate** to the program designed in Program 4 to perform the following task:
    a. Accepts two arguments , one of the structure **data** containing the present date and the second an integer that represents the number of days to be added to the present date.
    b. Adds the days to the present date and returns the structure containing the next that correctly.

```
#include <stdio.h>
#include <string.h>
struct date
{
   int day, month, year;
};
void readData(struct date *d)
{
   long int dateValue;
   printf("Enter date as a long integer (YYYYMMDD): ");
   scanf("%ld", &dateValue);

   d->year = (int)(dateValue / 10000);
   d->month = (int)((dateValue / 100) % 100);
   d->day = (int)(dateValue % 100);
}
int validateDate(struct date d)
{
   int daysInMonth[] = {0, 31, 28, 31, 30, 31, 30, 31, 31, 30, 31, 30, 31};

   if (d.year < 0 || d.month < 1 || d.month > 12 || d.day < 1)
      return 0;
   if ((d.year % 4 == 0 && d.year % 100 != 0) || (d.year % 400 == 0))
      daysInMonth[2] = 29;
   if (d.day > daysInMonth[d.month])
      return 0;
   return 1;
}
void printDate(struct date d)
{
   char *monthNames[] = {"", "January", "February", "March", "April", "May", "June",
"July", "August", "September", "October", "November", "December"};
   printf("%s %d, %d\n", monthNames[d.month], d.day, d.year);
}
```

```
struct date nextdate(struct date d, int daysToAdd)
{
   int daysInMonth[] = {0, 31, 28, 31, 30, 31, 30, 31, 31, 30, 31, 30, 31};
   while (daysToAdd > 0)
   {
      if ((d.year % 4 == 0 && d.year % 100 != 0) || (d.year % 400 == 0))
         daysInMonth[2] = 29;
      else
         daysInMonth[2] = 28;

      int daysRemainingInMonth = daysInMonth[d.month] - d.day;

      if (daysToAdd <= daysRemainingInMonth)
      {
         d.day += daysToAdd;
         daysToAdd = 0;
      }
      else
      {
         daysToAdd -= (daysRemainingInMonth + 1);
         d.day = 1;
         d.month++;

         if (d.month > 12)
         {
            d.month = 1;
            d.year++;
         }
      }
   }
   return d;
}
int main()
{
   struct date d;
   int n;

   readData(&d);
   if (validateDate(d))
   {
      printf("Current Date: ");
      printDate(d);

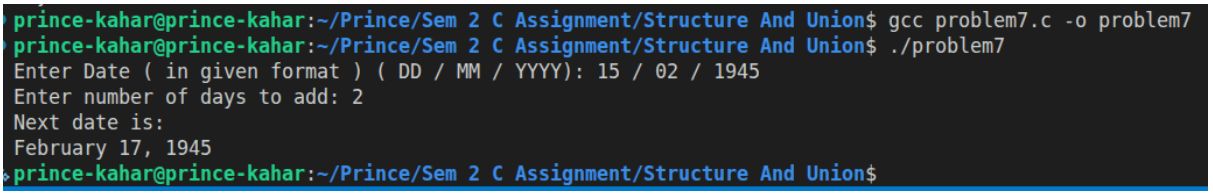
      printf("Enter number of days to add: ");
```

```
        scanf("%d", &n);

        d = nextdate(d, n);

        printf("Future Date: ");
        printDate(d);
    }
    else
        printf("Invalid date entered.\n");
    return 0;
}
```

Output :-



8. Use the **date** structure defined in Program 4 to store two dates. Develop a function that will take these two dates as input compares them
    a. it returns 1 , if the **date1** is earlier **date2**
    b. it return 0 , if **date1** is later date

```
#include <stdio.h>

struct date
{
   int day;
   int month;
   int year;
} d1, d2;

int leap_year(int year)
{
   if ((year % 4 == 0 && year % 100 != 0) || (year % 400 == 0))
      return 1;
   return 0;
}

int update_date(struct date d)
{
   int days_in_month[] = {0, 31, 28, 31, 30, 31, 30, 31, 31, 30, 31, 30, 31};

   if (d.year < 1 || d.day < 1 || d.day > days_in_month[d.month] || d.month < 1 || d.month > 12)
      return 0;

   if (d.month == 2 && leap_year(d.year))
      days_in_month[2] = 29;

   return 1;
}

int compare(struct date d1, struct date d2)
{
   if (d1.year < d2.year)
      return 1;
   else if (d1.year > d2.year)
      return 0;

   if (d1.month < d2.month)
      return 1;
```

```
    else if (d1.month > d2.month)
        return 0;

    if (d1.day < d2.day)
        return 1;
    else
        return 0;
}

int main()
{
    printf("Enter First Date ( in given format ) ( MM / DD / YYYY): ");
    if (scanf("%d / %d / %d", &d1.day, &d1.month, &d1.year) != 3)
    {
        printf("Error! Invalid date format ( Enter in MM / DD / YYYY format ) \n");
        return 0;
    } // check fromat

    if (!update_date(d1)) // check leap year
    {
        printf("Error! Invalid date entered.\n");
        return 0;
    }

    printf("Enter Second Date ( in given format ) ( MM / DD / YYYY): ");
    if (scanf("%d / %d / %d", &d2.day, &d2.month, &d2.year) != 3)
    {
        printf("Error! Invalid date format ( Enter in MM / DD / YYYY format ) \n");
        return 0;
    } // check fromat

    if (!update_date(d2)) // check leap year
    {
        printf("Error! Invalid date entered.\n");
        return 0;
    }

    compare(d1, d2) == 1 ? printf("Date1 is earlier than Date2\n") : printf("Date1 is later than
Date2\n");
    return 0;
}
```

Output :-

```
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Structure And Union$ gcc problem8.c -o problem8
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Structure And Union$ ./problem8
Enter First Date ( in given format ) ( MM / DD / YYYY): 11 / 06 / 2007
Enter Second Date ( in given format ) ( MM / DD / YYYY): 11 / 05 / 2007
Date1 is later than Date2
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Structure And Union$
```

9. Define a structure to represent a vector (a series of integer values) and write a modular program to perform the following task:
    a. To create a vector
    b. To modify the value of a given element
    c. To multiply by a scalar value
    d. To display the vector in the form (10,20,30,.....)

```
#include <stdio.h>

struct vector
{
   int arr[100];
   int size;
}v;

void create_vector(struct vector *v)
{
   printf("Enter size of vector: ");
   scanf("%d", &v->size);

   for (int i = 0; i < v->size; i++)
   {
      printf("Enter %d Element : ", i + 1);
      scanf("%d", &v->arr[i]);
   }
}

void modify_element(struct vector *v, int index, int new_value)
{
   index >= 0 && index < v->size ? v->arr[index] = new_value : printf("Invalid index.\n");
}

void multiply_by_scalar(struct vector *v, int scalar)
{
   for (int i = 0; i < v->size; i++)
      v->arr[i] = v->arr[i] * scalar;
}

void display_vector(struct vector v)
{
   printf("( ");
```

```
    for (int i = 0; i < v.size; i++)
    {
        printf("%d", v.arr[i]);
        if (i != v.size - 1)
            printf(", ");
    }
    printf(" )\n");
}

int main()
{
    int index, value, scalar;

    create_vector(&v);

    printf("Enter index to modify ( index start from 0 ): ");
    scanf("%d", &index);

    printf("Enter new value: ");
    scanf("%d", &value);
    modify_element(&v, index, value);

    printf("Enter scalar value: ");
    scanf("%d", &scalar);
    multiply_by_scalar(&v, scalar);

    printf("Final vector:\n");
    display_vector(v);

    return 0;
}
```

Output :-

```
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Structure And Union$ gcc problem9.c -o problem9
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Structure And Union$ ./problem9
Enter size of vector: 5
Enter 1 Element : 10
Enter 2 Element : 20
Enter 3 Element : 30
Enter 4 Element : 40
Enter 5 Element : 50
Enter index to modify ( index start from 0 ): 2
Enter new value: 5
Enter scalar value: 2
Final vector:
( 20, 40, 10, 80, 100 )
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Structure And Union$
```

10. Add a function to the program of Program 9 that accepts two vectors as input parameters and return the addition of two vectors.

```
#include <stdio.h>

struct vector
{
   int arr[100];
   int size;
} v1, v2, add;

void create_vector(struct vector *v, int size)
{
   for (int i = 0; i < size; i++)
   {
      printf("Enter %d Element : ", i + 1);
      scanf("%d", &v->arr[i]);
   }
}

int addition_of_vectors(struct vector *v1, struct vector *v2, struct vector *addition)
{
   if (v1->size != v2->size)
      return 0;

   addition->size = v1->size;

   for (int i = 0; i < v1->size; i++)
      addition->arr[i] = v1->arr[i] + v2->arr[i];
   return 1;
}

void display_vector(struct vector v)
{
   printf("( ");
   for (int i = 0; i < v.size; i++)
   {
      printf("%d", v.arr[i]);
      if (i != v.size - 1)
         printf(", ");
   }
   printf(" )\n");
}
```

```
int main()
{
   printf("Enter First Vector Size: ");
   scanf("%d", &v1.size);

   printf("Enter Second Vector Size: ");
   scanf("%d", &v2.size);

   if (v1.size != v2.size)
   {
      printf("\nVector addition not possible. Sizes are different.\n");
      return 0;
   }
   printf("\nEnter First Vector : \n");
   create_vector(&v1, v1.size);

   printf("\nEnter Second Vector : \n");
   create_vector(&v2, v2.size);

   addition_of_vectors(&v1, &v2, &add);
   printf("Addition of Vectors:\n   ");
   display_vector(v1);
   printf(" + ");
   display_vector(v2);
   printf("--------------------\n");
   printf(" = ");
   display_vector(add);
   return 0;
}
```

Output :-

```
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Structure And Union$ gcc problem10.c -o problem10
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Structure And Union$ ./problem10
Enter First Vector Size: 5
Enter Second Vector Size: 6

Vector addition not possible. Sizes are different.
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Structure And Union$ ./problem10
Enter First Vector Size: 2
Enter Second Vector Size: 2

Enter First Vector :
Enter 1 Element : 10
Enter 2 Element : 20

Enter Second Vector :
Enter 1 Element : 3
Enter 2 Element : 1
Addition of Vectors:
   ( 10, 20 )
 + ( 3, 1 )
--------------------
 = ( 13, 21 )
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Structure And Union$
```

11. Create two structures named **metric** and **British** which store the values of distances. The **metric** structure stores the values in metres and centimetres and the **British** structure stores the values in feet and inches. Write a program that reads values for the structure variables and adds values contained in one variable of **metric** to the contents of another variable of **British** . The program should display the result in the format of feet and inches or metres and centimetres as required.

```
#include <stdio.h>

struct metric
{
   int metre;
   int centimetre;
} m;

struct British
{
   int feet;
   int inch;
} b;

void add_into_metric(struct metric m, struct British b)
{
   float total_cm;

   total_cm = (m.metre * 100) + m.centimetre;
   total_cm += (b.feet * 30.48) + (b.inch * 2.54);

   m.metre = total_cm / 100;
   m.centimetre = (int)total_cm % 100;

   printf("Result in Metric System:\n");
   printf("%d metres %d centimetres\n", m.metre, m.centimetre);
}

void add_into_british(struct metric m, struct British b)
{
   float total_inches;

   total_inches = (m.metre * 100 + m.centimetre) / 2.54;
   total_inches += (b.feet * 12) + b.inch;
```

```
    b.feet = total_inches / 12;
    b.inch = (int)total_inches % 12;

    printf("Result in British System:\n");
    printf("%d feet %d inches\n", b.feet, b.inch);
}

int main()
{
    int choice;

    printf("Enter metric distance (metres & centimetres): ");
    scanf("%d %d", &m.metre, &m.centimetre);

    printf("Enter british distance (feet & inches): ");
    scanf("%d %d", &b.feet, &b.inch);

    printf("Display result in:\n");
    printf("1. Metric system\n");
    printf("2. British system\n");
    scanf("%d", &choice);

    if (choice == 1)
        add_into_metric(m, b);
    else
        add_into_british(m, b);
    return 0;
}
```

Output :-

```
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Structure And Union$ gcc problem11.c -o problem11
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Structure And Union$ ./problem11
Enter metric distance (metres & centimetres): 5 2
Enter british distance (feet & inches): 9 16
Display result in:
1. Metric system
2. British system
1
Result in Metric System:
8 metres 16 centimetres
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Structure And Union$
```

12. Define a structure named **census** with the following three members:
    a. A character array City [ ] to store names
    b. A long integer to store population of the city
    c. A float member to store the literacy level
    Write a program to do the following:
    a. To read details for 5 cities randomly using an array variable
    b. To sort the list alphabetically
    c. To sort the list based on literacy level
    d. To short the list based on population
    e. To display sorted lists

```
#include <stdio.h>
#include <string.h>

struct census
{
   char city[30];
   long int population;
   float literacy;
} city[5];

void input_data(struct census c[])
{
   for (int i = 0; i < 5; i++)
   {
      printf("\nEnter details of city %d\n", i + 1);
      printf("City name: ");
      scanf("%s", c[i].city);
      printf("Population ( in Lakhs ): ");
      scanf("%ld", &c[i].population);
      printf("Literacy level ( in Percentage ): ");
      scanf("%f", &c[i].literacy);
   }
}

void sort_by_name(struct census c[])
{
   struct census temp;
   for (int i = 0; i < 5 - 1; i++)
      for (int j = i + 1; j < 5; j++)
         if (strcmp(c[i].city, c[j].city) > 0)
         {
```

```
            temp = c[i];
            c[i] = c[j];
            c[j] = temp;
        }
}

void sort_by_literacy(struct census c[])
{
   struct census temp;
   for (int i = 0; i < 5 - 1; i++)
      for (int j = i + 1; j < 5; j++)
         if (c[i].literacy > c[j].literacy)
         {
            temp = c[i];
            c[i] = c[j];
            c[j] = temp;
         }
}

void sort_by_population(struct census c[])
{
   struct census temp;
   for (int i = 0; i < 5 - 1; i++)
      for (int j = i + 1; j < 5; j++)
         if (c[i].population > c[j].population)
         {
            temp = c[i];
            c[i] = c[j];
            c[j] = temp;
         }
}

void display(struct census c[])
{
   printf("\n%-15s %-15s %-15s\n", "City", "Population", "Literacy");
   for (int i = 0; i < 5; i++)
      printf("%-15s %-15ld %-15.2f\n", c[i].city, c[i].population, c[i].literacy);
}

int main()
{
   input_data(city);

   sort_by_name(city);
```

```
    printf("\nSorted Alphabetically (City Name):");
    display(city);

    sort_by_literacy(city);
    printf("\nSorted by Literacy Level:");
    display(city);

    sort_by_population(city);
    printf("\nSorted by Population:");
    display(city);

    return 0;
}
```

Output :-

```
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Structure And Union$ gcc problem12.c -o problem12
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Structure And Union$ ./problem12

Enter details of city 1
City name: Surat
Population ( in Lakhs ): 80
Literacy level ( in Percentage ): 87.89

Enter details of city 2
City name: Ahmedabad
Population ( in Lakhs ): 82
Literacy level ( in Percentage ): 88.29

Enter details of city 3
City name: Mumbai
Population ( in Lakhs ): 210
Literacy level ( in Percentage ): 89.73

Enter details of city 4
City name: Delhi
Population ( in Lakhs ): 320
Literacy level ( in Percentage ): 86.21

Enter details of city 5
City name: Banglore
Population ( in Lakhs ): 130
Literacy level ( in Percentage ): 88.48

Sorted Alphabetically (City Name):
City            Population      Literacy
Ahmedabad       82              88.29
Banglore        130             88.48
Delhi           320             86.21
Mumbai          210             89.73
Surat           80              87.89

Sorted by Literacy Level:
City            Population      Literacy
Delhi           320             86.21
Surat           80              87.89
Ahmedabad       82              88.29
Banglore        130             88.48
Mumbai          210             89.73

Sorted by Population:
City            Population      Literacy
Surat           80              87.89
Ahmedabad       82              88.29
Banglore        130             88.48
Mumbai          210             89.73
Delhi           320             86.21
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Structure And Union$
```

13. Define a structure that can describe a hotel . It should have members that include the name, address, grade, average room charge, and number of rooms.
    Write functions to perform the following operations:
    a. To print out hotels of a given grade in order of charges
    b. To print out hotels with room charges less than a given value

```
#include <stdio.h>
#include <string.h>

struct hotel
{
   char name[50];
   char address[100];
   int grade;
   float avg_room_charge;
   int no_of_rooms;
}h[5];

void print_by_grade(struct hotel h[], int n, int grade)
{
   for (int i = 0; i < n - 1; i++)
      for (int j = i + 1; j < n; j++)
         if (h[i].avg_room_charge > h[j].avg_room_charge)
         {
            struct hotel temp = h[i];
            h[i] = h[j];
            h[j] = temp;
         }

   printf("\nHotels of Grade %d in order of charges:\n", grade);
   printf("-----------------------------------------------------------------------\n");
   printf("%-25s %-20s %-10s %-10s\n", "NAME", "LOCATION", "CHARGE",
"ROOMS");
   for (int i = 0; i < n; i++)
      if (h[i].grade == grade)
         printf("%-25s %-20s %-10.2f %-10d\n",h[i].name, h[i].address,
h[i].avg_room_charge, h[i].no_of_rooms);
}

void print_by_charge(struct hotel h[], int n, float maxCharge)
{
```

```
    printf("\nHotels with room charge less than %.2f:\n", maxCharge);
    printf("------------------------------------------------------------------\n");
    for (int i = 0; i < n; i++)
        if (h[i].avg_room_charge < maxCharge)
            printf("Name: %s, Address: %s, Charge: %.2f\n",h[i].name, h[i].address,
h[i].avg_room_charge);
}

int main()
{
    int n = 5, grade;
    float charge;

    for (int i = 0; i < n; i++)
    {
        printf("\nEnter details of hotel %d:\n", i + 1);
        printf("Name: ");
        scanf(" %[^\n]", h[i].name);
        printf("Address: ");
        scanf(" %[^\n]", h[i].address);
        printf("Grade: ");
        scanf("%d", &h[i].grade);
        printf("Average Room Charge: ");
        scanf("%f", &h[i].avg_room_charge);
        printf("Number of Rooms: ");
        scanf("%d", &h[i].no_of_rooms);
    }

    printf("\nEnter grade to search: ");
    scanf("%d", &grade);
    print_by_grade(h, n, grade);

    printf("\nEnter maximum room charge: ");
    scanf("%f", &charge);
    print_by_charge(h, n, charge);

    return 0;
}
```

Output :-

```
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Structure And Union$ gcc problem13.c -o problem13
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Structure And Union$ ./problem13

Enter details of hotel 1:
Name: Taj Mahal Palace
Address: Mumbai
Grade: 5
Average Room Charge: 450.00
Number of Rooms: 560

Enter details of hotel 2:
Name: Taj Lake Palace
Address: Udaipur
Grade: 5
Average Room Charge: 650.00
Number of Rooms: 83

Enter details of hotel 3:
Name: ITC Grand Chola
Address: Chennai
Grade: 5
Average Room Charge: 250.00
Number of Rooms: 600

Enter details of hotel 4:
Name: Umaid Bhawan
Address: Jodhpur
Grade: 5
Average Room Charge: 800.00
Number of Rooms: 70

Enter details of hotel 5:
Name: Oberoi Amarvilas
Address: Agra
Grade: 5
Average Room Charge: 700.00
Number of Rooms: 102

Enter grade to search: 5

Hotels of Grade 5 in order of charges:
------------------------------------------------------------------------
NAME                    LOCATION            CHARGE     ROOMS
ITC Grand Chola         Chennai             250.00     600
Taj Mahal Palace        Mumbai              450.00     560
Taj Lake Palace         Udaipur             650.00     83
Oberoi Amarvilas        Agra                700.00     102
Umaid Bhawan            Jodhpur             800.00     70

Enter maximum room charge: 500

Hotels with room charge less than 500.00:
------------------------------------------------------------------------
Name: ITC Grand Chola, Address: Chennai, Charge: 250.00
Name: Taj Mahal Palace, Address: Mumbai, Charge: 450.00
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Structure And Union$
```

14. Define a structure called **cricket** that will describe the following information:

    player name
    team name
    batting average

    Using **cricket**, declare an array **player** with 50 elements and write a program to read the information about all the 50 players and print a team-wise list containing names of players with their batting average.

```
#include <stdio.h>
#include <string.h>
#define N 50

struct cricket
{
   char player_name[30], team_name[30];
   float betting_average;
}player[5]; // Using 5 for testing, change to 50 for final version

int main()
{
   int i, j;
   int printed[5] = {0}; // to track printed teams

   /* Input details */
   for (i = 0; i < 5; i++)
   {
      printf("\nEnter details of player %d\n", i + 1);

      printf("Player name: ");
      scanf("%[^\n]s", player[i].player_name);
      getchar();

      printf("Team name: ");
      scanf("%[^\n]s", player[i].team_name);

      printf("Batting average: ");
      scanf("%f", &player[i].betting_average);
      getchar();
   }
```

```
    printf("\nTEAM WISE PLAYER LIST\n");
    for (i = 0; i < 5; i++)
    {
        // Skip, If this player's team is already printed
        if (printed[i] == 1)
            continue;

        printf("\nTeam: %s\n", player[i].team_name);
        printf("----------------------------\n");
        printf("%-15s %-15s\n", "Player", "Average");

        for (j = 0; j < 5; j++)
            if (strcmp(player[i].team_name, player[j].team_name) == 0)
            {
                printf("%-15s %-15.2f\n", player[j].player_name, player[j].betting_average);
                printed[j] = 1;
            }
    }

    return 0;
}
```

Output :-

```
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Structure And Union$ gcc problem14.c -o problem14
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Structure And Union$ ./problem14

Enter details of player 1
Player name: Virat Kohli
Team name: India
Batting average: 58.67

Enter details of player 2
Player name: Glenn Maxwell
Team name: Australia
Batting average: 35.40

Enter details of player 3
Player name: Rohit Sharma
Team name: India
Batting average: 48.50

Enter details of player 4
Player name: Steve Smith
Team name: Australia
Batting average: 60.12

Enter details of player 5
Player name: Hardik Pandya
Team name: India
Batting average: 33.20

TEAM WISE PLAYER LIST

Team: India
---------------------------
Player          Average
Virat Kohli     58.67
Rohit Sharma    48.50
Hardik Pandya   33.20

Team: Australia
---------------------------
Player          Average
Glenn Maxwell   35.40
Steve Smith     60.12
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Structure And Union$
```

15. Design a structure **student_record** to contain name, date of birth, and total marks obtained. Use the **date** structure designed in Program 4 to represent the date of birth.
    Develop a program to read data for 10 students in a class and list them Rank-wise.

```
#include <stdio.h>
#define n 10

struct student_record
{
   char name[30];
   struct date
   {
      int day, month, year;
   } dob;
   float marks;
}s[n];

void input_student_record(struct student_record s[])
{
   for (int i = 0; i < n; i++)
   {
      printf("Enter details of student %d\n", i + 1);
      printf("Name: ");
      scanf("%[^\n]s", s[i].name);

      printf("Date of Birth (DD MM YYYY): ");
      scanf("%d %d %d", &s[i].dob.day, &s[i].dob.month, &s[i].dob.year);

      printf("Total Marks: ");
      scanf("%f", &s[i].marks);
      getchar();
      printf("\n");
   }
}

void sort_rankwise(struct student_record s[])
{
   for (int i = 0; i < n - 1; i++)
      for (int j = i + 1; j < n; j++)
         if (s[i].marks < s[j].marks)
         {
```

```
            struct student_record temp = s[i];
            s[i] = s[j];
            s[j] = temp;
        }
}

void display(struct student_record s[])
{
   printf("\nRANK-WISE STUDENT LIST\n");
   printf("-------------------------------------------------\n");
   printf("Rank  Name          DOB           Marks\n");
   printf("-------------------------------------------------\n");

   for (int i = 0; i < n; i++)
      printf("%-5d %-15s %02d/%02d/%04d      %-10.2f\n", i + 1, s[i].name, s[i].dob.day,
s[i].dob.month, s[i].dob.year, s[i].marks);
}

int main()
{
   input_student_record(s);
   sort_rankwise(s);
   display(s);

   return 0;
}
```

Output :-

```
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Structure And Union$ gcc problem15.c -o problem15
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Structure And Union$ ./problem15
Enter details of student 1
Name: Aarav Sharma
Date of Birth (DD MM YYYY): 14 05 2007
Total Marks: 258

Enter details of student 2
Name: Ishani Patel
Date of Birth (DD MM YYYY): 22 09 2007
Total Marks: 262

Enter details of student 3
Name: Rohan Das
Date of Birth (DD MM YYYY): 05 01 2008
Total Marks: 207

Enter details of student 4
Name: Ananya Iyer
Date of Birth (DD MM YYYY): 30 11 2007
Total Marks: 292

Enter details of student 5
Name: Kabir Malhotra
Date of Birth (DD MM YYYY): 12 03 2008
Total Marks: 227

Enter details of student 6
Name: Meera Reddy
Date of Birth (DD MM YYYY): 18 07 2007
Total Marks: 246

Enter details of student 7
Name: Sahil Kapoor
Date of Birth (DD MM YYYY): 09 02 2008
Total Marks: 180

Enter details of student 8
Name: Diya Mistri
Date of Birth (DD MM YYYY): 25 10 2007
Total Marks: 275

Enter details of student 9
Name: Aryan Verma
Date of Birth (DD MM YYYY): 11 12 2007
Total Marks: 230

Enter details of student 10
Name: Sana Khan
Date of Birth (DD MM YYYY): 03 06 2008
Total Marks: 271


RANK-WISE STUDENT LIST
-----------------------------------------------------
Rank  Name           DOB            Marks
-----------------------------------------------------
1     Ananya Iyer    30/11/2007     292.00
2     Diya Mistri    25/10/2007     275.00
3     Sana Khan      03/06/2008     271.00
4     Ishani Patel   22/09/2007     262.00
5     Aarav Sharma   14/05/2007     258.00
6     Meera Reddy    18/07/2007     246.00
7     Aryan Verma    11/12/2007     230.00
8     Kabir Malhotra 12/03/2008     227.00
9     Rohan Das      05/01/2008     207.00
10    Sahil Kapoor   09/02/2008     180.00
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Structure And Union$
```

16. Write a C program that prints the size of a structure data type.

```
#include <stdio.h>

struct student
{
   char name[50];
   int roll_no;
   float marks;
}s;

int main()
{
   printf("The size of the student is %lu bytes\n",sizeof(s));
   return 0;
}
```

Output :-

```
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Structure And Union$ gcc problem16.c -o problem16
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Structure And Union$ ./problem16
The size of the student is 60 bytes
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Structure And Union$
```

17. Write a C program that prints the size of a structure and union data type that have same number and type of members.

```
struct Student
{
   char name[50];
   int roll_no;
   float marks;
}S;

union student
{
   char name[50];
   int roll_no;
   float marks;
}s;

int main()
{
   printf("The size of the struct is %lu bytes\n", sizeof(S));
   printf("The size of the union is %lu bytes\n", sizeof(s));
   return 0;
}
```

Output :-

```
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Structure And Union$ gcc problem17.c -o problem17
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Structure And Union$ ./problem17
The size of the struct is 60 bytes
The size of the union is 52 bytes
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Structure And Union$
```

18. Write a C program for demonstrating operations on individual structure members using pointer notation.

```
#include <stdio.h>

struct student
{
   int roll_no;
   float marks;
}s,*ptr;

int main()
{
   ptr = &s;
   ptr->roll_no = 101;
   ptr->marks = 88.50;

   printf("Roll Number : %d\n", ptr->roll_no);
   printf("Marks       : %.2f\n", ptr->marks);

   return 0;
}
```

Output :-

```
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Structure And Union$ gcc problem18.c -o problem18
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Structure And Union$ ./problem18
Roll Number : 101
Marks       : 88.50
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Structure And Union$
```

User Defined Function

1. Write a function **exchange** to interchange the values of two variables, say **x** and **y**. Illustrate the use of this function, in a calling function. Assume that **x** and **y** are defined as global variables.

```
#include <stdio.h>

int a = 10, b = 20;

void exchange(void)
{
   b = a + b;
   a = b - a;
   b = b - a;
}

int main()
{
   printf("Before exchange: a = %d, b = %d\n", a, b);
   exchange();
   printf("After exchange: a = %d, b = %d\n", a, b);
   return 0;
}
```

Output :-

```
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/User Defined Function$ gcc problem1.c -o problem1
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/User Defined Function$ ./problem1
Before exchange: a = 10, b = 20
After exchange: a = 20, b = 10
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/User Defined Function$
```

2. Write a function **space(x)** that can be used to provide a space of x positions between two output numbers. Demonstrate its application.

```
#include <stdio.h>

void space(int x)
{
   for (int i = 0; i < x; i++)
      printf(" ");
}

int main()
{
   printf("123");
   space(3);
   printf("456\n");
   return 0;
}
```

Output :-

```
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/User Defined Function$ gcc problem2.c -o problem2
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/User Defined Function$ ./problem2
Enter the first number : 10
Enter the second number : 20
How many spaces you have include between two numbers : 5
10     20
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/User Defined Function$
```

3. Use recursive function calls to evaluate

$$f(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \text{.....}$$

```
#include <stdio.h>
#include <math.h>

int factorial(int n)
{
   if (n <= 1)
      return 1;
   else
      return n * factorial(n - 1);
}

float evaluate(int x, int power, int i, int terms)
// x = value of x, power = current power, i = current term number, terms = total number of
terms to evaluate
{
   if (power > terms * 2 - 1) // Highest power is 2 * terms - 1
      return 0;
   float term = pow(x, power) / factorial(power);
   if (i % 2 == 0)
      return -term + evaluate(x, power + 2, i + 1, terms); // if terms is even then negative
   else
      return term + evaluate(x, power + 2, i + 1, terms); // if terms is odd then positive
}

int main()
{
   int x, terms;

   printf("Enter the value of x : ");
   scanf("%d", &x) != 1;
   printf("Enter the number of terms to evaluate : ");
   if(scanf("%d", &terms) != 1)
   {
      printf("Error reading terms value\n");
      return 1;
   }

   printf("The value of f(x) is : ");
   printf("%f\n", evaluate(x, 1, 1, terms));
```

```
    return 0;
}
```

Output :-

```
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/User Defined Function$ gcc problem3.c -o problem3 -lm
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/User Defined Function$ ./problem3
Enter the value of x : 2
Enter the number of terms to evaluate : 5
The value of f(x) is : 0.909347
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/User Defined Function$
```

4. An n_order polynomial can be evaluated as follows:

$$P = (\ldots..(((a_0x + a_1)x + a_2)x + a_3)x + \ldots + a_n)$$

Write a function to evaluate the polynomial, using an array variable. Test it using a main program.

```
#include <stdio.h>

float evaluate_the_polynomial(float a[], int n, float x)
{
   float p = a[0];
   for (int i = 1; i <= n; i++)
      p = p * x + a[i];
   return p;
}

int main()
{
   int n;
   float x;
   printf("Enter polynomial (N) : ");
   scanf("%d", &n);

   float coeff[n + 1];
   printf("Enter %d Coefficicents : \n", n);
   for (int i = 0; i <= n; i++)
   {
      printf("Enter coefficients[%d] : ", i);
      scanf("%f", &coeff[i]);
   }

   printf("Enter The Value of X ( that is multiply ) : ");
   scanf("%f", &x);

   printf("Answer P(%f) is %f \n", x, evaluate_the_polynomial(coeff, n, x));
   return 0;
}
```

Output :-

```
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/User Defined Function$ gcc problem4.c -o problem4
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/User Defined Function$ ./problem4
Enter polynomial (N) : 2
Enter 2 Coefficicents :
Enter coefficients[0] : 2
Enter coefficients[1] : 3
Enter coefficients[2] : 5
Enter The Value of X ( that is multiply ) : 2
Answer P(2.000000) is 19.000000
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/User Defined Function$
```

5. The Fibonacci numbers are defined recursively as follows:

$$f_1 = 1$$
$$f_2 = 1$$
$$f_n = f_{n-1} + f_{n-2}, n > 2$$

Write a function that will generate and print the first n Fibonacci numbers. Test the function for n = 5, 10, and 15.

```
#include <stdio.h>

int fibonacci(int n)
{
   if (n <= 0)
   {
      printf("Please enter a positive integer.");
      return -1;
   }
   int a = 0, b = 1;
   for (int i = 0; i < n; i++)
   {
      int next = a + b;
      printf("%d ", next);
      a = b;
      b = next;
   }
   return 0;
}

int main()
{
   printf("The first 5 Fibonacci numbers are: ");
   fibonacci(5);
   printf("\n");
   printf("The first 10 Fibonacci numbers are: ");
   fibonacci(10);
   printf("\n");
   printf("The first 15 Fibonacci numbers are: ");
   fibonacci(15);
   printf("\n");
   return 0;
}
```

Output :-

```
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/User Defined Function$ gcc problem5.c -o problem5
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/User Defined Function$ ./problem5
The first 5 Fibonacci numbers are: 1 2 3 5 8
The first 10 Fibonacci numbers are: 1 2 3 5 8 13 21 34 55 89
The first 15 Fibonacci numbers are: 1 2 3 5 8 13 21 34 55 89 144 233 377 610 987
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/User Defined Function$
```

6. Write a function that will round a floating-point number to an indicated decimal place.

```
#include <stdio.h>
#include <math.h>

void rounded(float num, int places)
{
   num = round(num * pow(10, places)) / pow(10, places);
   printf("Rounded value: %.*f\n", places, num);
}

int main()
{
   float num;
   int decimal;
   printf("Enter number to round: ");
   scanf("%f", &num);
   printf("Enter number of decimal places to round to: ");
   scanf("%d", &decimal);
   rounded(num, decimal);
   return 0;
}
```

Output :-

```
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/User Defined Function$ gcc problem6.c -o problem6 -lm
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/User Defined Function$ ./problem6
Enter number to round: 17.457
Enter number of decimal places to round to: 2
Rounded value: 17.46
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/User Defined Function$
```

7. Write a function **prime** that returns 1 if its argument is a prime number and returns zero otherwise.

```
#include <stdio.h>

int prime(int n)
{
   if (n <= 1)
      return 0;

   for (int i = 2; i * i <= n; i++)
      if (n % i == 0)
         return 0;
   return 1;
}

int main()
{
   int n;
   printf("Enter number to check prime: ");
   scanf("%d", &n);

   if (prime(n))
      printf("%d is prime\n", n);
   else
      printf("%d is not prime\n", n);
   return 0;
}
```

Output :-

```
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/User Defined Function$ gcc problem7.c -o problem7
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/User Defined Function$ ./problem7
Enter number to check prime: 5
5 is prime
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/User Defined Function$ ./problem7
Enter number to check prime: 10
10 is not prime
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/User Defined Function$
```

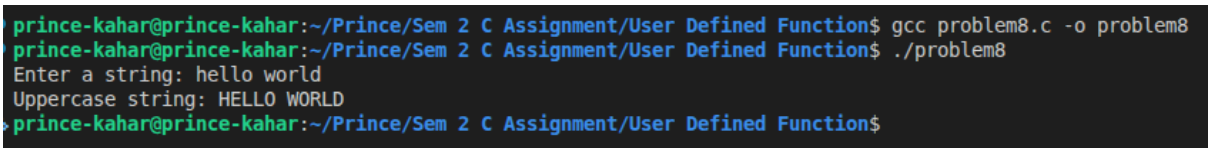
8. Write a function that will scan a character string passed as an argument and convert all lowercase characters into their uppercase equivalents.

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

void convert_to_uppercase(char *str)
{
   for (int i = 0; str[i] != '\0'; i++)
      str[i] = toupper(str[i]);
}

int main()
{
   char str[100];
   printf("Enter a string: ");
   fgets(str, sizeof(str), stdin);
   convert_to_uppercase(str);
   printf("Uppercase string: %s", str);
   return 0;
}
```

Output :-



9. Develop a top_down modular program to implement a calculator. The program should request the user to input two numbers and display one of the following as per the desire of the user:
    a. Sum of the numbers
    b. Difference of the numbers
    c. Product of the numbers
    d. Division of the numbers

    Provide separate functions for performing various tasks such as reading, calculating and displaying. Calculating module should call second level modules to perform the individual mathematical operations. The main function should have only function calls.

```
#include <stdio.h>
#define sum(a, b) ((a + b))
#define difference(a, b) ((a - b))
#define product(a, b) ((a * b))
#define division(a, b) ((b != 0) ? (a / b) : (printf("Error: Division by zero!\n"), 0))

void reading(float *a, float *b)
{
   printf("Enter two numbers: ");
   scanf("%f %f", a, b);
}

char operation()
{
   char choice;
   printf("\nChoose operation:\n");
   printf("A. Sum\n");
   printf("B. Difference\n");
   printf("C. Product\n");
   printf("D. Division\n");
   printf("Enter your choice (A/B/C/D): ");
   scanf(" %c", &choice);
   return choice;
}

void displaying(float result)
{
   printf("Result: %.2f\n", result);
}
```

```
void calculating(float a, float b, char operation)
{
   float result;
   switch (operation)
   {
   case 'A':
   case 'a':
      result = sum(a, b);
      break;
   case 'B':
   case 'b':
      result = difference(a, b);
      break;
   case 'C':
   case 'c':
      result = product(a, b);
      break;
   case 'D':
   case 'd':
      result = division(a, b);
      break;
   default:
      printf("Error! Invalid operation!\n");
      return;
   }
   displaying(result);
}

int main()
{
   float num1, num2;
   reading(&num1, &num2);
   calculating(num1, num2, operation());
   return 0;
}
```

Output :-

```
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/User Defined Function$ gcc problem9.c -o problem9
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/User Defined Function$ ./problem9
Enter two numbers: 5 10

Choose operation:
A. Sum
B. Difference
C. Product
D. Division
Enter your choice (A/B/C/D): A
Result: 15.00
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/User Defined Function$
```

10. Develop a modular interactive program using functions that reads the values of three sides of a triangle and displays either its area or its perimeter as per the request of the user. Given the three sides a, b and c.

$$Perimeter = a + b + c$$

$$Area = \sqrt[2]{s * (s - a) * (s - b) * (s - c)}$$

Where $s = \frac{(a + b + c)}{2}$

```
#include <stdio.h>
#include <math.h>

void read_sides(float *a, float *b, float *c)
{
   printf("Enter first side: ");
   scanf("%f", a);
   printf("Enter second side: ");
   scanf("%f", b);
   printf("Enter third side: ");
   scanf("%f", c);
}

int valid_triangle(float a, float b, float c)
{
   if ((a + b > c) && (a + c > b) && (b + c > a))
      return 1;
   return 0;
}

float perimeter(float a, float b, float c)
{
   return a + b + c;
}

float area(float a, float b, float c)
{
   float s = (a + b + c) / 2;
   return sqrt(s * (s - a) * (s - b) * (s - c));
}

int main()
{
```

```
    float a, b, c;
    char operation;

    read_sides(&a, &b, &c);
    if(!valid_triangle(a,b,c))
    {
        printf("Error! The given sides do not form a valid triangle.\n");
        return 1;
    }
    printf("Enter one of the followings: ");
    printf("\n\tA. Perimeter of the triangle ");
    printf("\n\tB. Area of the triangle ");
    printf("\n");

    scanf(" %c", &operation);
    if(operation == 'A' || operation == 'a')
        printf("Result -> %.2f\n", perimeter(a, b, c));
    else if(operation == 'B' || operation == 'b')
        printf("Result -> %.2f\n", area(a, b, c));
    else
        printf("Invalid operation\n");
    return 0;
}
```

Output :-

```
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/User Defined Function$ gcc problem10.c -o problem10 -lm
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/User Defined Function$ ./problem10
Enter first side: 2
Enter second side: 3
Enter third side: 4
Enter one of the followings:
        A. Perimeter of the triangle
        B. Area of the triangle
a
Result -> 9.00
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/User Defined Function$
```

11. Write a function that can be called to find the largest element of an m by n matrix.

```
#include <stdio.h>

int largest_element(int m, int n, int matrix[m][n])
{
   int largest_element = matrix[0][0];
   for (int i = 0; i < m; i++)
      for (int j = 0; j < n; j++)
         if (matrix[i][j] > largest_element)
            largest_element = matrix[i][j];
   return largest_element;
}

int main()
{
   int m, n;
   printf("Enter the number of rows and columns of the matrix: ");
   scanf("%d %d", &m, &n);

   int matrix[m][n];
   printf("Enter the elements of the matrix: \n");
   for (int i = 0; i < m; i++)
      for (int j = 0; j < n; j++)
      {
         printf("Enter Element [%d][%d] : ", i, j);
         scanf("%d", &matrix[i][j]);
      }
   printf("The largest element in the matrix is: %d\n", largest_element(m, n, matrix));

   return 0;
}
```

Output :-

```
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/User Defined Function$ gcc problem11.c -o problem11
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/User Defined Function$ ./problem11
Enter the number of rows and columns of the matrix: 2 3
Enter the elements of the matrix:
Enter Element [0][0] : 1
Enter Element [0][1] : 2
Enter Element [0][2] : 5
Enter Element [1][0] : 1
Enter Element [1][1] : 9
Enter Element [1][2] : 4
The largest element in the matrix is: 9
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/User Defined Function$
```

12. Write a function that can be called to compute the product of two matrices of size m by n and n by m. The main function provides the values for m and n and two matrices.

```
#include <stdio.h>

void element(int *element, int row, int col)
{
   for (int i = 0; i < row; i++)
      for (int j = 0; j < col; j++)
      {
         printf("Enter Element [%d][%d] : ", i, j);
         scanf("%d", &element[i * col + j]);
      }
}

void matrices_product(int m, int n, int matrix1[][n], int matrix2[][m], int result[][m])
{
   for (int i = 0; i < m; i++)
      for (int j = 0; j < m; j++)
      {
         int sum = 0;
         for (int k = 0; k < n; k++)
            sum += matrix1[i][k] * matrix2[k][j];
         result[i][j] = sum;
      }
}

int main()
{
   int rows1, cols1, rows2, cols2;

   while (1)
   {
      printf("Enter the number of rows and columns for Matrix 1: ");
      scanf("%d %d", &rows1, &cols1);
      printf("Enter the number of rows and columns for Matrix 2: ");
      scanf("%d %d", &rows2, &cols2);

      if (cols1 != rows2)
      {
         printf("Error: matrix 1 column cannot equal matrix 2 rows\n\n");
         continue;
```

```
        }

        int matrix1[rows1][cols1], matrix2[rows2][cols2], result[rows1][cols2];

        printf("Enter the elements of the first matrix:\n");
        element(&matrix1[0][0], rows1, cols1);

        printf("Enter the elements of the second matrix:\n");
        element(&matrix2[0][0], rows2, cols2);

        matrices_product(rows1, cols1, matrix1, matrix2, result);
        printf("Product of the matrices:\n");
        for (int i = 0; i < rows1; i++)
        {
            for (int j = 0; j < cols2; j++)
                printf("%d ", result[i][j]);
            printf("\n");
        }
        return 0;
    }
}
```

Output :-

```
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/User Defined Function$ gcc problem12.c -o problem12
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/User Defined Function$ ./problem12
Enter the number of rows and columns for Matrix 1: 2 3
Enter the number of rows and columns for Matrix 2: 2 3
Error: matrix 1 column cannot equal matrix 2 rows

Enter the number of rows and columns for Matrix 1: 2 2
Enter the number of rows and columns for Matrix 2: 2 2
Enter the elements of the first matrix:
Enter Element [0][0] : 1
Enter Element [0][1] : 2
Enter Element [1][0] : 3
Enter Element [1][1] : 4
Enter the elements of the second matrix:
Enter Element [0][0] : 5
Enter Element [0][1] : 6
Enter Element [1][0] : 7
Enter Element [1][1] : 8
Product of the matrices:
19 22
43 50
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/User Defined Function$
```

13. Design and code an interactive modular program that will use functions to a matrix of m by n size, compute column averages and row averages, and then print the entire matrix with averages shown in respective rows and columns.

```
#include <stdio.h>

void average_of_row_and_col(int m, int n, int matrix[][n], float row_averages[], float
column_averages[])
{
   for (int i = 0; i < m; i++)
   {
      row_averages[i] = 0;
      for (int j = 0; j < n; j++)
         row_averages[i] += matrix[i][j];
      row_averages[i] /= n;
   }

   for (int j = 0; j < n; j++)
   {
      column_averages[j] = 0;
      for (int i = 0; i < m; i++)
         column_averages[j] += matrix[i][j];
      column_averages[j] /= m;
   }
}

void display(int m, int n, int matrix[][n], float row_averages[], float column_averages[])
{
   printf("\nMatrix : \n");
   for (int i = 0; i < m; i++)
   {
      for (int j = 0; j < n; j++)
         printf("%d  ", matrix[i][j]);
      printf("\n");
   }

   printf("\nRow Average : ");
   for (int i = 0; i < m; i++)
      printf("%.2f  ", row_averages[i]);

   printf("\nColumn Averages : ");
   for (int j = 0; j < n; j++)
```

```
        printf("%.2f  ", column_averages[j]);
    printf("\n");
}

int main()
{
    int m, n;

    printf("Enter the size of the matrix (m and n): ");
    scanf("%d %d", &m, &n);

    int matrix[m][n];
    float row_averages[m];
    float column_averages[n];

    printf("\nEnter the elements of the matrix:\n");
    for (int i = 0; i < m; i++)
        for (int j = 0; j < n; j++)
        {
            printf("Enter Element [%d][%d]: ", i, j);
            scanf("%d", &matrix[i][j]);
        }

    average_of_row_and_col(m, n, matrix, row_averages, column_averages);
    display(m, n, matrix, row_averages, column_averages);

    return 0;
}
```

Output :-

```
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/User Defined Function$ gcc problem13.c -o problem13
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/User Defined Function$ ./problem13
Enter the size of the matrix (m and n): 2 2

Enter the elements of the matrix:
Enter Element [0][0]: 10
Enter Element [0][1]: 19
Enter Element [1][0]: 43
Enter Element [1][1]: 61

Matrix :
10  19
43  61

Row Average : 14.50  52.00
Column Averages : 26.50  40.00
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/User Defined Function$
```

14. Develop a top-down modular program that will perform the following tasks:
    a. Read two integer arrays with unsorted elements.
    b. Sort them in ascending order
    c. Merge the sorted arrays
    d. Print the sorted list

    Use functions for carrying out each of the above tasks. The main function should have only function calls.

```
#include <stdio.h>

void input_array(int array[], int size)
{
   printf("\nEnter Unsorted Element : \n");
   for (int i = 0; i < size; i++)
   {
      printf("Enter Element %d: ", i + 1);
      scanf("%d", &array[i]);
   }
   printf("\n");
}

void sort_array(int array[], int size)
{
   for (int i = 0; i < size; i++)
   {
      int min = array[i], min_index = i;
      for (int j = i + 1; j < size; j++)
         if (array[j] < min)
         {
            min = array[j];
            min_index = j;
         }
      int temp = array[i];
      array[i] = array[min_index];
      array[min_index] = temp;
   }
}

void merge_arrays(int array1[], int array2[], int result[], int m, int n)
{
   int i, j;
```

```
    for (i = 0; i < m; i++)
        result[i] = array1[i];
    for (j = 0; j < n; j++)
        result[i + j] = array2[j];
}

void print_array(int array[], int size)
{
    printf("Array: \n");
    for (int i = 0; i < size; i++)
        printf("%d ", array[i]);
    printf("\n");
}

void operations()
{
    int m, n;
    printf("Enter size of array 1 and array 2 : ");
    scanf("%d %d", &m, &n);

    int array1[m], array2[n];

    printf("Input for array 1:\n");
    input_array(array1, m);

    printf("Input for array 2:\n");
    input_array(array2, n);

    sort_array(array1, m);
    sort_array(array2, n);

    int result[m + n];
    merge_arrays(array1, array2, result, m, n);
    sort_array(result, m + n);

    print_array(result, m + n);
}

int main()
{
    operations();
    return 0;
}
```

Output :-

```
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/User Defined Function$ gcc problem14.c -o problem14
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/User Defined Function$ ./problem14
Enter size of array 1 and array 2 : 3 5
Input for array 1:

Enter Unsorted Element :
Enter Element 1: 60
Enter Element 2: 10
Enter Element 3: 94

Input for array 2:

Enter Unsorted Element :
Enter Element 1: 16
Enter Element 2: 40
Enter Element 3: 51
Enter Element 4: 37
Enter Element 5: 94

Array:
10 16 37 40 51 60 94 94
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/User Defined Function$
```

15. Develop your own functions for performing following operations on strings:
    a. Copying one string to another
    b. Comparing two strings
    c. Adding a string to the end of another string

    Write a driver program to test your functions.

```
#include <stdio.h>
#include <string.h>

void copy_string(char dest[], char src[])
{
   int i;
   for (i = 0; src[i] != '\0'; i++)
      dest[i] = src[i];
   dest[i] = '\0';
}

int compare_string(char str1[], char str2[])
{
   int i;
   for (i = 0; str1[i] == str2[i]; i++)
      if (str1[i] == '\0')
         return 0;
   return str1[i] - str2[i];
}

void concatenate_string(char dest[], const char src[])
{
   int i, j;
   for (i = 0; dest[i] != '\0'; i++); // Move to the end of the destination string
   for (j = 0; src[j] != '\0'; j++) // Copy characters from source string to destination string
   {
      dest[i] = src[j];
      i++;
   }
   dest[i] = '\0';
}

int main()
{
   char str1[100], str2[100], copy_str[100];
```

```
    printf("Enter your first string: ");
    scanf("%[^\n]", str1);

    printf("Enter your second string: ");
    scanf(" %[^\n]", str2);

    // Testing copyString function
    copy_string(copy_str, str1);
    printf("\nCopied first string: %s\n", copy_str);

    // Testing stringCompare function
    compare_string(str1, str2) == 0 ? printf("The strings are equal\n") : printf("The string are
different\n");

    // Testing stringConcatenate function
    concatenate_string(str1, str2);
    printf("Concatenated string: %s\n", str1);

    return 0;
}
```

Output :-

```
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/User Defined Function$ gcc problem15.c -o problem15
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/User Defined Function$ ./problem15
Enter your first string: How are you ?
Enter your second string: I am fine

Copied first string: How are you ?
The string are different
Concatenated string: How are you ? I am fine
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/User Defined Function$
```

16. Write a program that invokes a function called **find( )** to perform the following tasks:
    a. Receives a character array and a single character.
    b. Returns 1 if the specified character is found in the array, 0 otherwise.

```
#include <stdio.h>

int find(char array[], char character)
{
   for (int i = 0; array[i] != '\0'; i++)
      if (array[i] == character)
         return 1;
   return 0;
}

int main()
{
   char str[100], find_character;

   printf("Enter a String: ");
   scanf("%[^\n]s", str);

   printf("Enter a character to search for: ");
   scanf(" %c", &find_character);

   if (find(str, find_character) == 1)
      printf("The character '%c' is found in the array.\n", find_character);
   else
      printf("The character '%c' is not found in the array.\n", find_character);
   return 0;
}
```

Output :-

```
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/User Defined Function$ gcc problem16.c -o problem16
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/User Defined Function$ ./problem16
Enter a String: Hello! I am good
Enter a character to search for: o
The character 'o' is found in the array.
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/User Defined Function$
```

17. Design a function **locate ( )** that takes two character arrays **s1** and **s2** and one integer value **m** as parameters and inserts the string **s2** into **s1** immediately after the index **m** .
Write a program to test the function using a real-life situation.

```
#include <stdio.h>
#include <string.h>

char* locate(char s1[], char s2[], int m)
{
   char temp[200];
   int i, j = 0;

   for (i = 0; i <= m; i++)
      temp[j++] = s1[i]; // copy characters from s1 to temp until index m

   for (i = 0; s2[i] != '\0'; i++)
      temp[j++] = s2[i]; // copy characters from s2 to temp

   for (i = m + 1; s1[i] != '\0'; i++)
      temp[j++] = s1[i]; // copy remaining characters from s1 to temp

   temp[j] = '\0';
   strcpy(s1, temp); // copy modified string back to s1

   return s1;
}

int main()
{
   char str1[200], str2[100];
   int m;

   printf("Enter the first string (str1): ");
   scanf("%[^\n]s", str1);
   printf("Enter the second string (str2): ");
   scanf(" %[^\n]s", str2);

   printf("Enter the index (m) after which to insert str2 into str1: ");
   scanf("%d", &m);

   locate(str1, str2, m);
   printf("The modified string is: %s\n", str1);
```

```
    return 0;
}
```

Output :-

```
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/User Defined Function$ gcc problem17.c -o problem17
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/User Defined Function$ ./problem17
Enter the first string (str1): Hello I am Fine
Enter the second string (str2): World
Enter the index (m) after which to insert str2 into str1: 5
The modified string is: Hello World I am Fine
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/User Defined Function$
```

18. Write a function that takes an integer parameter **m** representing the month number of the year and returns the corresponding name of the month.

```
#include <stdio.h>

char *month_name(int m)
{
   static char months[12][10] = {"January", "February", "March", "April", "May", "June",
"July", "August", "September", "October", "November", "December"};

   if (m < 1 || m > 12)
   {
      printf("Error! Invalid month\n");
      return NULL;
   }
   return months[m - 1];
}

int main()
{
   int m;
   printf("Enter month number (1-12): ");
   if (scanf("%d", &m) != 1)
   /*
   Here Check condition !
   If the input is not an integer, scanf will return a value different from 1.
   And go to if block.
   */
   {
      printf("Invalid input.\n");
      return 1;
   }

   char *name = month_name(m);
   name != NULL ? printf("Month Name is %s\n", name) : printf("Error! Can't Retrive
Month Name .\n");
   return 0;
}
```

Output :-

```
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/User Defined Function$ gcc problem18.c -o problem18
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/User Defined Function$ ./problem18
Enter month number (1-12): 3
Month Name is March
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/User Defined Function$
```

19. In preparing the calendar for a year we need to know whether that particular year is leap year or not. Design a function **leap( )** that receives the year as a parameter and returns an appropriate message.

```
#include <stdio.h>

void leap(int year)
{
   if ((year % 4 == 0 && year % 100 != 0) || (year % 400 == 0))
      printf("%d is a leap year.\n", year);
   else
      printf("%d is not a leap year.\n", year);
}

int main()
{
   int year;
   printf("Enter a year: ");
   if (scanf("%d", &year) != 1)
   {
      printf("Invalid input.\n");
      return 1;
   }

   leap(year);
   return 0;
}
```

Output :-

```
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/User Defined Function$ gcc problem19.c -o problem19
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/User Defined Function$ ./problem19
Enter a year: 2002
2002 is not a leap year.
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/User Defined Function$
```

20. Write a function that receives a floating point value x and returns it as a value rounded to two nearest decimal places.

```
#include <stdio.h>

void rounded(float x)
{
   printf("Rounded value: %.2f\n", x);
}

int main(void)
{
   float x;
   printf("Enter a floating point value: ");
   if (scanf("%f", &x) != 1)
   {
      printf("Invalid input.\n");
      return 1;
   }

   rounded(x);
   return 0;
}
```

Output :-

```
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/User Defined Function$ gcc problem20.c -o problem20
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/User Defined Function$ ./problem20
Enter a floating point value: 123.4567
Rounded value: 123.46
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/User Defined Function$
```

Pointers

1. Write a program using pointers to read in an array of integers and print its elements in reverse order.

```
#include <stdio.h>
#include <stdlib.h>

int main()
{
   int size, i;
   printf("Enter Array Size : ");
   scanf("%d", &size);

   int *arr = (int *)malloc(size * sizeof(int));
   int *ptr = arr;

   for (i = 0; i < size; i++)
   {
      printf("Enter %d Element : ", i + 1);
      scanf("%d", (ptr + i));
   }

   printf("\nReverse Order : ");
   for (i = size - 1; i >= 0; i--)
      printf("%d\t", *(ptr + i));
   printf("\n");

   free(arr);
   return 0;
}
```

Output :-

```
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Pointers$ gcc problem1.c -o problem1
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Pointers$ ./problem1
Enter Array Size : 5
Enter 1 Element : 60
Enter 2 Element : 42
Enter 3 Element : 91
Enter 4 Element : 16
Enter 5 Element : 81

Reverse Order : 81      16      91      42      60
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Pointers$
```

2. We know that the roots of a quadratic equation of the form

$$ax^2 + bx + c = 0$$

are given by the following equations :

$$x_1 = \frac{-b + \sqrt[2]{(b^2 - 4ac)}}{2a}$$

$$x_2 = \frac{-b - \sqrt[2]{(b^2 - 4ac)}}{2a}$$

Write a function to calculate the roots. The function must use two pointer parameters, one to receive the coefficients a, b, and c, and the other to send the roots to the calling function.

```
#include <stdio.h>
#include <math.h>

float* quadratic_equation(float *coefficients, float *roots)
{
   float a = *coefficients, b = *(coefficients + 1), c = *(coefficients + 2);
   float discriminant = b * b - 4 * a * c;
   if (discriminant >= 0)
   {
      *(roots) = (-b + sqrt(discriminant)) / (2 * a);
      *(roots + 1) = (-b - sqrt(discriminant)) / (2 * a);
   }
   return roots;
}

int main()
{
   float coefficients[3], roots[2];
   printf("Enter Coefficients of a, b and c : ");
   scanf("%f %f %f", &coefficients[0], &coefficients[1], &coefficients[2]);

   quadratic_equation(coefficients, roots);
   printf("These Quadratic Equation Roots Are %.3f and %.3f", roots[0], roots[1]);
   printf("\n");
   return 0;
}
```

Output :-

```
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Pointers$ gcc problem2.c -o problem2 -lm
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Pointers$ ./problem2
Enter Coefficients of a, b and c : 1 -5 6
These Quadratic Equation Roots Are 3.000 and 2.000
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Pointers$
```

3. Write a function that receives a sorted array of integers and an integer value, and inserts the value in its correct place.

```
#include <stdio.h>
#include <stdlib.h>

int check_sorted(int *arr, int size)
{
   for (int i = 0; i < size - 1; i++)
      if (arr[i] > arr[i + 1])
         return 0;
   return 1;
}

int insert_element(int *arr, int *size, int newElement)
{
   int i;
   for (i = *size - 1; i >= 0 && *(arr + i) > newElement; i--)
      *(arr + i + 1) = *(arr + i);
   *(arr + i + 1) = newElement;
   (*size)++;
   return 1;
}

int main()
{
   int size, i, new_element;
   printf("Enter Array Size : ");
   scanf("%d", &size);
   int *arr = (int *)malloc((size + 1) * sizeof(int));

   while (1)
   {
      printf("\nEnter Sorted Element \n");
      for (i = 0; i < size; i++)
      {
         printf("Enter %d Element : ", i + 1);
         scanf("%d", &arr[i]);
      }
      if (!check_sorted(arr, size))
      {
         printf("\nError! Please Enter Sorted Element\n");
         continue;
```

```
        }
        break;
    }

    printf("\nEnter a Value to insert : ");
    scanf("%d", &new_element);

    insert_element(arr, &size, new_element);
    printf("Updated Array : \n");
    for (i = 0; i < size; i++)
        printf("%d ", arr[i]);
    printf("\n");

    free(arr);
    return 0;
}
```

Output :-

```
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Pointers$ gcc problem3.c -o problem3
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Pointers$ ./problem3
Enter Array Size : 5

Enter Sorted Element
Enter 1 Element : 10
Enter 2 Element : 20
Enter 3 Element : 30
Enter 4 Element : 50
Enter 5 Element : 40

Error! Please Enter Sorted Element

Enter Sorted Element
Enter 1 Element : 10
Enter 2 Element : 20
Enter 3 Element : 30
Enter 4 Element : 40
Enter 5 Element : 50

Enter a Value to insert : 24
Updated Array :
10 20 24 30 40 50
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Pointers$
```

4. Write a function that pointers to add two matrices and to return the resultant matrix to the calling function

```
#include <stdio.h>

void input(int *arr, int row, int col, int n)
{
   for (int i = 0; i < row; i++)
      for (int j = 0; j < col; j++)
      {
         printf("Enter array%d[%d][%d] : ", n, i, j);
         scanf("%d", &arr[i * col + j]);
      }
}

void add_two_matrices(int *ptr, int *ptr2, int *ans, int row, int col)
{
   for (int i = 0; i < row * col; i++)
      *(ans + i) = *(ptr + i) + *(ptr2 + i);
}

int main()
{
   while (1)
   {
      int i, j, row1, col1, row2, col2;
      printf("Enter Matrix 1 Row and Column : ");
      scanf("%d %d", &row1, &col1);
      printf("Enter Matrix 2 Row and Column : ");
      scanf("%d %d", &row2, &col2);

      if (row1 == row2 && col1 == col2)
      {
         int matrix1[row1][col1], matrix2[row2][col2], add[row1][col1];
         input(*matrix1, row1, col1, 1);

         printf("\n");
         input(*matrix2, row2, col2, 2);

         add_two_matrices(*matrix1, *matrix2, *add, row1, col1);
         printf("Addition of the given Matrices is \n");

         for (i = 0; i < 2; i++)
```

```
        {
            for (j = 0; j < 2; j++)
                printf("%d\t", add[i][j]);
            printf("\n");
        }
    }
    else
    {
        printf("Error! Matrix Order is different \n\n");
        continue;
    }
    break;
  }
  return 0;
}
```

Output :-

```
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Pointers$ gcc problem4.c -o problem4
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Pointers$ ./problem4
Enter Matrix 1 Row and Column : 2 2
Enter Matrix 2 Row and Column : 3 3
Error! Matrix Order is different

Enter Matrix 1 Row and Column : 2 2
Enter Matrix 2 Row and Column : 2 3
Error! Matrix Order is different

Enter Matrix 1 Row and Column : 2 2
Enter Matrix 2 Row and Column : 2 2
Enter array1[0][0] : 10
Enter array1[0][1] : 20
Enter array1[1][0] : 30
Enter array1[1][1] : 40

Enter array2[0][0] : 12
Enter array2[0][1] : 13
Enter array2[1][0] : 14
Enter array2[1][1] : 16
Addition of the given Matrices is
22      33
44      56
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Pointers$
```

5. Using pointers, write a function that receives a character string and a character as argument and deletes all occurrences of this character in the string . The function should return the corrected string with no holes.

```
#include <stdio.h>

void remove_char(char *str, char removable_char)
{
   char *src = str, *dest = str;
   while (*src != '\0')
   {
      if (*src != removable_char)
      {
         *dest = *src;
         dest++;
      }
      src++;
   }
   *dest = '\0';
}

int main()
{
   char str[100], removeableChar;
   printf("Enter Your String: ");
   scanf("%[^\n]s", str);

   printf("Which Character You Have to Remove form the String : ");
   scanf(" %c", &removeableChar);

   remove_char(str, removeableChar);

   printf("%s\n", str);
   return 0;
}
```

Output :-

```
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Pointers$ gcc problem5.c -o problem5
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Pointers$ ./problem5
Enter Your String: Hello World
Which Character You Have to Remove form the String : l
Heo Word
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Pointers$
```

6. Write a function **day_name** that receives a number n and returns a pointer to a character string containing the name of corresponding day. The day name should be kept in a static table of character strings local to the function.

```
#include<stdio.h>

void day_name(int n)
{
   static char *days[] = {"Invalid Day", "Monday", "Tuesday", "Wednesday", "Thursday",
"Friday", "Saturday", "Sunday"};
   printf("The Day is : %s\n",(n >= 1 && n<=7) ? days[n] : days[0]);
}

int main()
{
   int n;
   printf("Enter Number between 1 to 7 : ");
   scanf("%d", &n);

   day_name(n);
   return 0;
}
```

Output :-

```
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Pointers$ gcc problem6.c -o problem6
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Pointers$ ./problem6
Enter Number between 1 to 7 : 7
The Day is : Sunday
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Pointers$
```

7. Write a program to read in an array of names and to sort them in alphabetical order. Use **sort** function that receives pointers to the function **strcmp** and **swap.sort** in turn should call these functions via the pointers.

```
#include <stdio.h>
#include <string.h>
#include <stdlib.h>

int i;

void swap(char **a, char **b)
{
   char *temp = *a;
   *a = *b;
   *b = temp;
}

void sort(char *name[], int n, int (*cmp)(const char *, const char *))
{
   for (i = 0; i < n - 1; i++)
      for (int j = i + 1; j < n; j++)
         if (strcmp(name[i], name[j]) > 0)
            swap(&name[i], &name[j]);
}

int main()
{
   char **names = (char **)malloc(3 * sizeof(char *));

   printf("Enter 3 Name : ");
   for (i = 0; i < 3; i++)
   {
      names[i] = (char *)malloc(50 * sizeof(char));
      scanf("%s", names[i]);
   }

   sort(names, 3, strcmp);

   printf("\nsorted names :\n");
   for (i = 0; i < 3; i++)
      printf("%s\n", names[i]);

   free(names);
```

```
    return 0;
}
```

Output :-

```
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Pointers$ gcc problem7.c -o problem7
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Pointers$ ./problem7
Enter 3 Name : Hello World Earth

sorted names :
Earth
Hello
World
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Pointers$
```

8. Given an array of sorted list of integer numbers, Write a function to search for a particular item, using the method of binary search. And also show how this function may be used in a program. use pointers and pointer arithmetic.

```
#include <stdio.h>

int check_sorted(int *arr, int size)
{
   for (int i = 0; i < size - 1; i++)
      if (arr[i] > arr[i + 1])
         return 0;
   return 1;
}

int binarySearch(int *arr, int size, int find)
{
   int *low = arr;
   int *high = arr + size - 1;

   while (low <= high)
   {
      int *mid = low + (high - low) / 2;

      if (*mid == find)
         return (int)(mid - arr);
      else if (*mid < find)
         low = mid + 1;
      else
         high = mid - 1;
   }
   return -1;
}

int main()
{
   int size, i, arr[100], key;

   printf("Enter Array Size : ");
   scanf("%d", &size);

   while (1)
   {
```

```
        printf("\nEnter Sorted Element \n");
        for (i = 0; i < size; i++)
        {
            printf("Enter %d Element : ", i + 1);
            scanf("%d", &arr[i]);
        }
        if (!check_sorted(arr, size))
        {
            printf("\nError! Please Enter Sorted Element\n");
            continue;
        }
        break;
    }

    printf("\nWhich Element You Have To Find : ");
    scanf("%d", &key);
    int result = binarySearch(arr, size, key);

    if (result != -1)
        printf("\nElement %d found at Position : %d\n", key, result + 1);
    else
        printf("\nElement not found.\n");
    return 0;
}
```

Output :-

```
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Pointers$ gcc problem8.c -o problem8
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Pointers$ ./problem8
Enter Array Size : 3

Enter Sorted Element
Enter 1 Element : 10
Enter 2 Element : 30
Enter 3 Element : 20

Error! Please Enter Sorted Element

Enter Sorted Element
Enter 1 Element : 10
Enter 2 Element : 20
Enter 3 Element : 30

Which Element You Have To Find : 20

Element 20 found at Position : 2
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Pointers$
```

9. Write a function (using a pointer parameter) that reverses the elements of a given array.

```
#include <stdio.h>
#include <stdlib.h>

void swap(int *left, int *right)
{
   int temp = *left;
   *left = *right;
   *right = temp;
}

void reverses_array(int *arr, int n)
{
   int *left = arr;          // starting point
   int *right = arr + n - 1; // ending point
   while (left < right)
   {
      swap(left, right);
      left++;
      right--;
   }
}

int main()
{
   int size, i;
   printf("Enter Array Size : ");
   scanf("%d", &size);
   int *arr = (int *)malloc((size + 1) * sizeof(int));

   for (i = 0; i < size; i++)
   {
      printf("Enter %d Element : ", i + 1);
      scanf("%d", &arr[i]);
   }

   reverses_array(arr, size);
   for (i = 0; i < size; i++)
      printf("%d\t", arr[i]);
   printf("\n");
```

```
    free(arr);
    return 0;
}
```

Output :-

```
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Pointers$ gcc problem9.c -o problem9
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Pointers$ ./problem9
Enter Array Size : 3
Enter 1 Element : 10
Enter 2 Element : 30
Enter 3 Element : 20
20      30      10
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Pointers$
```

10. Write a function (using pointer parameters) that compares two integer arrays to see whether they are identical. The function returns 1 if they are identical, 0 otherwise.

```
#include <stdio.h>

void input(int *arr, int size, int n)
{
   for (int i = 0; i < size; i++)
   {
      printf("Enter array%d[%d]: ", n, i);
      scanf("%d", &arr[i]);
   }
}

int comapare_two_integer(int *arr1, int *arr2, int arr1size, int arr2size)
{
   if (arr1size != arr2size)
      return 0;
   else
      for (int i = 0; i < arr1size; i++)
         if (*(arr1 + i) != *(arr2 + i))
            return 0;
}

int main()
{
   int arr1_size, arr2_size, i, j, arr1[100], arr2[100];
   printf("Enter Array 1 Size : ");
   scanf("%d", &arr1_size);

   printf("Enter Array 2 Size : ");
   scanf("%d", &arr2_size);
   if (arr1_size == arr2_size)
   {
      input(arr1, arr1_size, 1);
      printf("\n");
      input(arr2, arr2_size, 1);

      int b = comapare_two_integer(arr1, arr2, arr1_size, arr2_size);
      if (b == 0)
         printf("Both Array are Diffrenet\n");
      else
```

```
            printf("Both Array are Identical\n");
    }
    else
        printf("Both Array Size are Differenet\n");

    return 0;
}
```

Output :-

```
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Pointers$ gcc problem10.c -o problem10
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Pointers$ ./problem10
Enter Array 1 Size : 2
Enter Array 2 Size : 3
Both Array Size are Differenet
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Pointers$ ./problem10
Enter Array 1 Size : 2
Enter Array 2 Size : 2
Enter array1[0]: 1
Enter array1[1]: 2

Enter array1[0]: 1
Enter array1[1]: 6
Both Array are Diffrenet
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Pointers$ ./problem10
Enter Array 1 Size : 2
Enter Array 2 Size : 2
Enter array1[0]: 1
Enter array1[1]: 2

Enter array1[0]: 1
Enter array1[1]: 2
Both Array are Identical
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Pointers$
```

11. Write a C program to insert a substring into another string by using function and pointers.

```
#include <stdio.h>
#include <string.h>

void insertSubString(char *main_str, char *ptr_sub_str, int position)
{
   char temp[200];
   char *ptr_main_str = main_str;
   char *ptr_temp = temp;

   for (int i = 0; i < position; i++)
      *ptr_temp++ = *ptr_main_str++; // copy from main string to position to temp

   while (*ptr_sub_str)
      *ptr_temp++ = *ptr_sub_str++; // copy from sub string to temp where we left above
loop

   while (*ptr_main_str)
      *ptr_temp++ = *ptr_main_str++; // copy from remaining main str to temp where we left
above loop

   *ptr_temp = '\0';
   strcpy(main_str, temp); // copy string from temp to main string
}

int main()
{
   char str[100], sub_str[100];
   int position;

   printf("Enter Your String : ");
   scanf("%[^\n]s", str);
   getchar();
   printf("Enter Your SubString : ");
   scanf("%[^\n]s", sub_str);

   printf("Enter the Position to Insert at ( 1 to %lu ) : ",strlen(str)  );
   scanf("%d", &position);

   printf("\nOriginal String : %s\n", str);
   insertSubString(str, sub_str, position);
```

```
    printf("\nAfter Insertion SubString : %s\n", str);

    return 0;
}
```

Output :-

```
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Pointers$ gcc problem11.c -o problem11
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Pointers$ ./problem11
Enter Your String : Hello I am Fine
Enter Your SubString : World
Enter the Position to Insert at ( 1 to 15 ) : 6

Original String : Hello I am Fine

After Insertion SubString : Hello World I am Fine
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Pointers$
```

12. Write a C program that uses the pointer increment operations to demonstrate the scale factor.

```
#include <stdio.h>

int main()
{
   int integer = 0, *ptr_integer = &integer;
   char charater = 'a', *ptr_character = &charater;
   double doubLe = 0.0, *ptr_doubLe = &doubLe;

   printf("Int Pointer     :   %p -> After Increment Operations : %p (Difference: %ld bytes)\n", ptr_integer, ptr_integer + 1, (long)(ptr_integer + 1) - (long)ptr_integer);
   printf("Char Pointer    :   %p -> After Increment Operations : %p (Difference: %ld bytes)\n", ptr_character, ptr_character + 1, (long)(ptr_character + 1) - (long)ptr_character);
   printf("Double Pointer  :   %p -> After Increment Operations : %p (Difference: %ld bytes)\n", ptr_doubLe, ptr_doubLe + 1, (long)(ptr_doubLe + 1) - (long)ptr_doubLe);

   return 0;
}
```

Output :-

```
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Pointers$ gcc problem12.c -o problem12
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Pointers$ ./problem12
Int Pointer     :   0x7fff16c4a2b4 -> After Increment Operations : 0x7fff16c4a2b8 (Difference: 4 bytes)
Char Pointer    :   0x7fff16c4a2b3 -> After Increment Operations : 0x7fff16c4a2b4 (Difference: 1 bytes)
Double Pointer  :   0x7fff16c4a2b8 -> After Increment Operations : 0x7fff16c4a2c0 (Difference: 8 bytes)
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Pointers$
```

13. Write a C program that displays the addresses and values pointed by an array of pointers.

```
#include <stdio.h>

int main()
{
   int size, i, arr[100], *ptr[100];
   printf("Enter Array Size : ");
   scanf("%d", &size);
   for (i = 0; i < size; i++)
   {
      printf("Enter %d Value : ", i + 1);
      scanf("%d", &arr[i]);
   }
   printf("\n");

   for (i = 0; i < size; i++)
      ptr[i] = &arr[i]; // Array of pointers
   for (i = 0; i < size; i++)
      printf("Pointer index %d is at address %p, points to value %d\n", i, &ptr[i], *ptr[i]);

   return 0;
}
```

Output :-

```
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Pointers$ gcc problem13.c -o problem13
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Pointers$ ./problem13
Enter Array Size : 5
Enter 1 Value : 10
Enter 2 Value : 20
Enter 3 Value : 40
Enter 4 Value : 50
Enter 5 Value : 60

Pointer index 0 is at address 0x7ffef9baf3c0, points to value 10
Pointer index 1 is at address 0x7ffef9baf3c8, points to value 20
Pointer index 2 is at address 0x7ffef9baf3d0, points to value 40
Pointer index 3 is at address 0x7ffef9baf3d8, points to value 50
Pointer index 4 is at address 0x7ffef9baf3e0, points to value 60
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Pointers$
```

14. Write a C program that uses pointers to demonstrate initialization of structure members.

```
#include <stdio.h>

struct Student
{
   int roll;
   float marks;
} s1, *ptr;

int main()
{
   ptr = &s1;

   printf("Enter Student Roll No : ");
   scanf("%d", &ptr->roll);
   printf("Enter Student Marks : ");
   scanf("%f", &ptr->marks);

   printf("Student Roll: %d \n\tMarks: %.2f\n", ptr->roll, ptr->marks);
   return 0;
}
```

Output :-

```
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Pointers$ gcc problem14.c -o problem14
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Pointers$ ./problem14
Enter Student Roll No : 101
Enter Student Marks : 50.00
Student Roll: 101
        Marks: 50.00
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/Pointers$
```

File Management

1. Write a program to copy the contents of one into another.

```
#include <stdio.h>

int main()
{
   FILE *fp1, *fp2;

   char ch;
   char f1[50], f2[50];

   printf("Enter first file name: ");
   scanf("%s", f1);
   printf("Enter second file name: ");
   scanf("%s", f2);

   fp1 = fopen(f1, "r");
   if (fp1 == NULL)
   {
      printf("\nError! %s is Not Open\n", f1);
      printf("\nIf you wish to create %s ( Y/n ) :- ", f1);
      char ch;
      scanf(" %c", &ch);
      if (ch == 'Y' || ch == 'y')
      {
         while (getchar() != '\n');

         fp1 = fopen(f1, "w+");
         printf("Enter text for %s (end with ~ and then enter) : ", f1);
         while ((ch = getchar()) != '~') // Read characters until '~' is encountered
            fputc(ch, fp1);

         rewind(fp1);
      }
      else
         return 1;
   }
   else
      printf("%s is Opened Successfully\n", f1);

   fp2 = fopen(f2, "w");
```

```
    if (fp2 == NULL)
    {
        printf("\nError! %s is Not Open\n", f2);
        return 1;
    }
    else
        printf("\n%s is Opened Successfully\n", f2);

    while ((ch = fgetc(fp1)) != EOF)
        fputc(ch, fp2);

    printf("File copied successfully.\n");

    fclose(fp1);
    fclose(fp2);
    return 0;
}
```

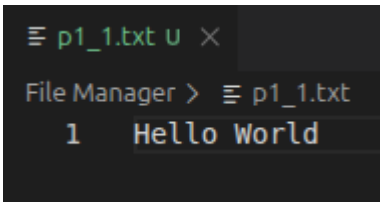
Output :-

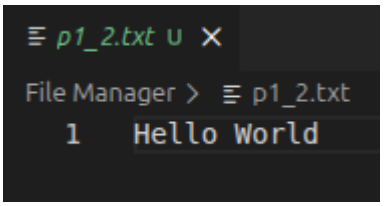
```
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/File Manager$ gcc problem1.c -o problem1
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/File Manager$ ./problem1
Enter first file name: p1_1.txt
Enter second file name: p1_2.txt

Error! p1_1.txt is Not Open

If you wish to create p1_1.txt ( Y/n ) :- y
Enter text for p1_1.txt (end with ~ and then enter) : Hello World~

p1_2.txt is Opened Successfully
File copied successfully.
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/File Manager$
```





2. Two files DATA1 and DATA2 contain sorted lists of integers. Write a program to produce a third file DATA which holds a single sorted, merged list of these two lists. Use command line arguments to specify the file names.

```
#include <stdio.h>

void create_and_fill_file(const char *filename)
{
   FILE *fp = fopen(filename, "w");
   if (fp == NULL)
   {
      printf("Fatal Error: Could not create %s\n", filename);
      return ;
   }

   int count, current, previous;

   printf("\n---------- Creating %s ----------\n", filename);
   printf("How many integers do you want to enter for %s? ", filename);
   while (scanf("%d", &count) != 1 || count <= 0)
   {
      printf("[!] WARNING: Enter a valid positive integer for count: ");
      while (getchar() != '\n'); // Clear Buffer
   }

   int first_valid = 0;
   while (!first_valid)
   {
      printf("Enter 1 integer: ");
      if (scanf("%d", &previous) == 1)
      {
         fprintf(fp, "%d\n", previous);
         first_valid = 1;
      }
      else
      {
         printf("\n[!] WARNING: Only integers are allowed! Try again.\n");
         while (getchar() != '\n'); // clear Bufffer
      }
   }

   for (int i = 0; i < count-1; i++)
```

```
    {
        printf("Enter %d integer (must be >= %d): ", i + 2, previous);
        if (scanf("%d", &current) != 1) // Check if input is an integer
        {
            printf("[!] WARNING: Character/String/Special_Charater not allowed. Enter an
integer! Try Again.\n");
            while (getchar() != '\n'); // Clear Buffer
            i--;  // Repeat this step
            continue;
        }

        if (current < previous) // Check if input is sorted
        {
            printf("\n[!] WARNING: Numbers must be in SORTED order (Greater than %d)! Try
again.\n", previous);
            i--;
            continue;
        }

        fprintf(fp, "%d\n", current);
        previous = current;
    }
    fclose(fp);
    printf("[SUCCESS] %s created with sorted data.\n", filename);
}

int main(int argc, char *argv[])
{
    FILE *fp1, *fp2, *fp3;
    char ch;
    int num1, num2;

    if (argc != 4)
    {
        printf("Usage: %s %s %s %s\n", argv[0], argv[1], argv[2], argv[3]);
        return 1;
    }

    fp1 = fopen(argv[1], "r");
    if (fp1 == NULL)
    {
        printf("\n%s not found. If you wish to create it? (Y/n): ", argv[1]);
        scanf(" %c", &ch);
        if (ch == 'y' || ch == 'Y')
```

```
    {
        create_and_fill_file(argv[1]);
        fp1 = fopen(argv[1], "r");
    }
    else
        return 1;
}

fp2 = fopen(argv[2], "r");
if (fp2 == NULL)
{
    printf("\n%s not found. If you wish to create? (Y/n): ", argv[2]);
    scanf(" %c", &ch);
    if (ch == 'y' || ch == 'Y')
    {
        create_and_fill_file(argv[2]);
        fp2 = fopen(argv[2], "r");
    }
    else
    {
        fclose(fp1);
        return 1;
    }
}

fp3 = fopen(argv[3], "w");

int result1 = fscanf(fp1, "%d", &num1);
int result2 = fscanf(fp2, "%d", &num2);

while (result1 == 1 && result2 == 1)
{
    if (num1 < num2)
    {
        fprintf(fp3, "%d\n", num1);
        result1 = fscanf(fp1, "%d", &num1);
    }
    else
    {
        fprintf(fp3, "%d\n", num2);
        result2 = fscanf(fp2, "%d", &num2);
    }
} // it is stop when one of the file is end of file
while (result1 == 1) // write the remaining numbers of the file which is not end of file
```

```
    {
        fprintf(fp3, "%d\n", num1);
        result1 = fscanf(fp1, "%d", &num1);
    }
    while (result2 == 1) // write the remaining numbers of the file which is not end of file
    {
        fprintf(fp3, "%d\n", num2);
        result2 = fscanf(fp2, "%d", &num2);
    }

    fclose(fp1);
    fclose(fp2);
    fclose(fp3);
    printf("\n[DONE] Files merged successfully into %s\n", argv[3]);
    return 0;
}
```

Output :-

```
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/File Manager$ gcc problem2.c -o problem2
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/File Manager$ ./problem2 DATA1 DATA2 DATA

DATA1 not found. If you wish to create it? (Y/n): y

------------------ Creating DATA1 ------------------
How many integers do you want to enter for DATA1? 5
Enter 1 integer: 1
Enter 2 integer (must be >= 1): 2
Enter 3 integer (must be >= 2): 60
Enter 4 integer (must be >= 60): 40

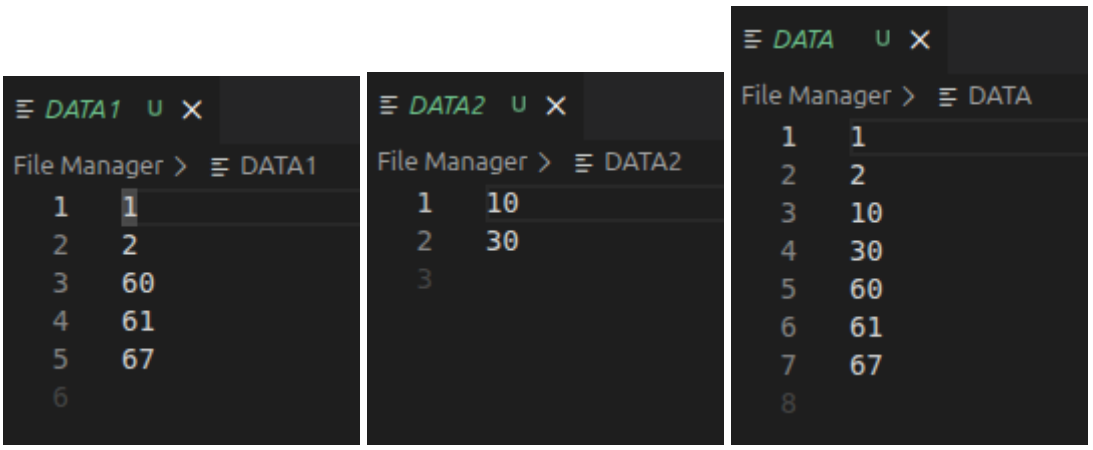
[!] WARNING: Numbers must be in SORTED order (Greater than 60)! Try again.
Enter 4 integer (must be >= 60): 20

[!] WARNING: Numbers must be in SORTED order (Greater than 60)! Try again.
Enter 4 integer (must be >= 60): 61
Enter 5 integer (must be >= 61): 67
[SUCCESS] DATA1 created with sorted data.

DATA2 not found. If you wish to create? (Y/n): y

------------------ Creating DATA2 ------------------
How many integers do you want to enter for DATA2? 2
Enter 1 integer: 10
Enter 2 integer (must be >= 10): 30
[SUCCESS] DATA2 created with sorted data.

prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/File Manager$
```


DATA1
File Manager > DATA1
1 1
2 2
3 60
4 61
5 67
6
DATA2
File Manager > DATA2
1 10
2 30
3
DATA
File Manager > DATA
1 1
2 2
3 10
4 30
5 60
6 61
7 67
8

3. Write a program that compares two files and returns 0 if they are equal and 1 if they are not.

```
#include <stdio.h>

int main()
{
   FILE *fp1, *fp2;

   int num1, num2;
   char ch1, ch2;
   char f1[50], f2[50];

   printf("Enter first file name: ");
   scanf("%s", f1);

   printf("Enter second file name: ");
   scanf("%s", f2);
   getchar();

   fp1 = fopen(f1, "w+");
   fp2 = fopen(f2, "w+");

   printf("Enter text for %s file (end with ~ and enter):\n", f1);
   while ((ch1 = getchar()) != '~')
      fputc(ch1, fp1);
   getchar();

   printf("Enter text for %s file (end with ~ and enter):\n", f2);
   while ((ch2 = getchar()) != '~')
      fputc(ch2, fp2);

   rewind(fp1);
   rewind(fp2);

   while (1)
   {
      ch1 = fgetc(fp1);
      ch2 = fgetc(fp2);

      if (ch1 != ch2)
      {
         fclose(fp1);
```

```
            fclose(fp2);
            printf("Files are not equal.\n");
            return 1; // Files are not equal
            /* give instruction to compiler that is a error in the program */
        }

        if (ch1 == EOF && ch2 == EOF) // Both Files are reached end point
            break;
    }

    printf("Files are equal.\n");
    fclose(fp1);
    fclose(fp2);

    return 0;
}
```

Output :-

```
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/File Manager$ gcc problem3.c -o problem3
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/File Manager$ ./problem3
Enter first file name: p3_1.txt
Enter second file name: p3_2.txt
Enter text for p3_1.txt file (end with ~ and enter):
Hello World~
Enter text for p3_2.txt file (end with ~ and enter):
Hello~
Files are not equal.
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/File Manager$ ./problem3
Enter first file name: p3_1.txt
Enter second file name: p3_2.txt
Enter text for p3_1.txt file (end with ~ and enter):
Hello World~
Enter text for p3_2.txt file (end with ~ and enter):
Hello World~
Files are equal.
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/File Manager$
```

4. Write a program that appends one file at the end of another.

```
#include <stdio.h>

int main()
{
   FILE *fp1, *fp2;
   char ch1, ch2;
   char f1[50], f2[50];

   printf("Enter first file name: ");
   scanf("%s", f1);
   printf("Enter second file name: ");
   scanf("%s", f2);
   getchar();

   fp1 = fopen(f1, "w");
   fp2 = fopen(f2, "w+");

   printf("Enter text for %s file (end with ~):\n", f1);
   while ((ch1 = getchar()) != '~')
      fputc(ch1, fp1);
   getchar();

   printf("Enter text for %s file (end with ~):\n", f2);
   while ((ch2 = getchar()) != '~')
      fputc(ch2, fp2);

   fclose(fp1);
   rewind(fp2);
   fp1 = fopen(f1, "a");

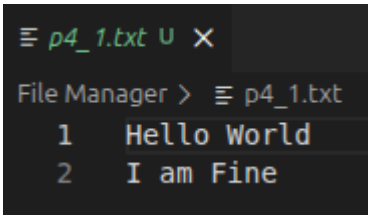
   while ((ch2 = fgetc(fp2)) != EOF)
      fputc(ch2, fp1);

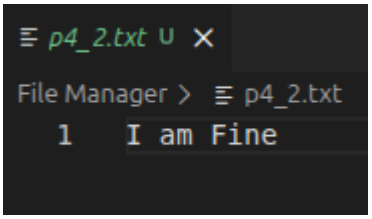
   fclose(fp1);
   fclose(fp2);

   printf("File appended successfully\n");
   return 0;
}
```

Output :-

```
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/File Manager$ gcc problem4.c -o problem4
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/File Manager$ ./problem4
Enter first file name: p4_1.txt
Enter second file name: p4_2.txt
Enter text for p4_1.txt file (end with ~):
Hello World
~
Enter text for p4_2.txt file (end with ~):
I am Fine~
File appended successfully
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/File Manager$
```





5. Write a program that reads a file containing integers and appends at its end the sum of all the integers.

```
#include <stdio.h>

int main()
{
   FILE *fp1;

   int num, sum = 0;
   char ch;
   char f1[50];

   printf("Enter file name: ");
   scanf("%s", f1);
   getchar();

   fp1 = fopen(f1, "w+");

   printf("Enter integer text for %s file (end with ~):\n", f1);
   while ((ch = getchar()) != '~')
      if (ch >= '0' && ch <= '9' || ch == '~' || ch == ' ') // allow only digits , ~ , and space
         fputc(ch, fp1);

   getchar();
   rewind(fp1);

   while (fscanf(fp1, "%d", &num) != EOF) // Read integer and sum
      sum += num;

   fclose(fp1);

   fp1 = fopen(f1, "a"); // open in append mode to add sum at the end
   if (fp1 == NULL)
   {
      printf("File is Not Open\n");
      return 1;
   }
   else
      printf("File is Opened Successfully in Append Mode\n");

   fprintf(fp1, "\nSum = %d", sum); // Add Sum at the end of file
   fclose(fp1);
```

```
    printf("Sum appended successfully\n");
    return 0;
}
```

Output :-

```
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/File Manager$ gcc problem5.c -o problem5
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/File Manager$ ./problem5
Enter file name: p5.txt
Enter integer text for p5.txt file (end with ~):
Hello 5 World 7
~
File is Opened Successfully in Append Mode
Sum appended successfully
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/File Manager$
```

p5.txt U

File Manager > p5.txt

```
1   5  7
2   Sum = 12
```

6. Write a program that prompts the user for two files, one containing a line of text known as source file and other, an empty file known as target file and then copies the contents of source file into target file.
   Modify the program so that a specified character is deleted from the source file as it is copied to the target file.

```
#include <stdio.h>

int main()
{
   FILE *src, *tgt;
   char ch1, ch2;

   src = fopen("source", "w+");

   printf("Enter text for source file (end with ~ and then enter ):\n");
   while ((ch1 = getchar()) != '~')
      fputc(ch1, src);

   rewind(src);
   tgt = fopen("target", "w");

   printf("Enter character to delete: ");
   scanf(" %c", &ch2); // Read the character to delete

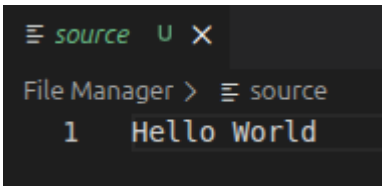
   while ((ch1 = fgetc(src)) != EOF)
   {
      if (ch1 != ch2) // Skip the specified character
         fputc(ch1, tgt);
   }

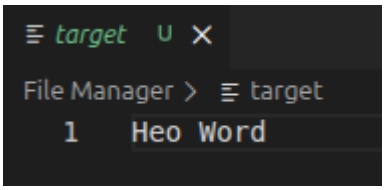
   fclose(src);
   fclose(tgt);

   printf("File copied successfully\n");
   return 0;
}
```

Output :-

```
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/File Manager$ gcc problem6.c -o problem6
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/File Manager$ ./problem6
Enter text for source file (end with ~ and then enter ):
Hello World~
Enter character to delete: l
File copied successfully
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/File Manager$
```





7. Write a program that requests for a file name and an integer, known as offset value. The program then reads the file starting from the location specified by the offset value and prints the contents on the screen.

```
#include <stdio.h>
#include <stdlib.h>

#define max_lines 1000
#define max_len 256

int main()
{
   FILE *fp;

   char filename[100];
   char lines[max_lines][max_len];
   int offset, count = 0, i;

   printf("Enter file name: ");
   scanf("%s", filename);

   fp = fopen(filename, "r");
   if (fp == NULL)
   {
      printf("File is Not Open\n");
      return 1;
   }
   else
      printf("File is Opened Successfully\n");

   printf("Enter offset value (must be integer): ");

   while (scanf("%d", &offset) != 1)
   {
      printf("\n[!] Warning Enter Only Integer Number \n");
      printf("Please Enter Integer Offset Value : ");
      while(getchar() != '\n');
   }

   while (fgets(lines[count], max_len, fp) != NULL) // Read all lines into array
   {
      count++;
      if (count >= max_lines)
```

```
            break;
    }
    fclose(fp);

    if (count == 0)
    {
        printf("Error: File is empty.\n");
        return 1;
    }
    /* Positive offset: skip lines from beginning */
    if (offset >= 0)
    {
        if (offset >= count)
        {
            printf("Error: Offset exceeds number of lines in file.\n");
            return 1;
        }
        for (i = offset; i < count; i++)
            printf("%s", lines[i]);
    }
    else
    {
        offset = -offset; // convert negative offset to the positive
        if (offset > count)
        {
            printf("Error: Offset exceeds number of lines.\n");
            return 1;
        }
        for (i = count - offset; i < count; i++)
            printf("%s", lines[i]);
    }
    return 0;
}
```

Output :-

```
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/File Manager$ gcc problem7.c -o problem7
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/File Manager$ ./problem7
Enter file name: offset.txt
File is Opened Successfully
Enter offset value (must be integer): vf

[!] Warning Enter Only Integer Number
Please Enter Integer Offset Value : 4
5. File Manager
6. Pre - Processor
7. File Manager program
8. Problem no 7prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/File Manager$
```

File Manager > offset.txt

```
1. Sem 2 Syllabus
2. Structure and Union
3. User Defined Function
4. Pointer
5. File Manager
6. Pre - Processor
7. File Manager program
8. Problem no 7
```

8. Write a program to create a sequential file that could store details about five products. Details include product code, cost and number of items available and are provided through keyboard.

```
#include <stdio.h>

struct product
{
   int code;
   float cost;
   int quantity;
}p;

int main()
{
   FILE *fp;

   int i,n;

   fp = fopen("product", "w");

   if (fp == NULL)
   {
      printf("File is Not Open\n");
      return 1;
   }
   else
      printf("File is Opened Successfully\n");

   fprintf(fp, "Code\t\t\tCost\t\t\tQuantity\n");

   printf("How many Product Details You have to Enter : ");
   scanf("%d",&n);

   for (i = 1; i <= n; i++)
   {
      printf("\nEnter details of product %d\n", i);

      printf("Product code: ");
      scanf("%d", &p.code);
      printf("Cost: ");
      scanf("%f", &p.cost);
      printf("Number of items available: ");
```

```
        scanf("%d", &p.quantity);

        fprintf(fp, "%-15d %-15.2f %-15d\n", p.code, p.cost, p.quantity);
    }

    fclose(fp);

    printf("\nProduct details stored successfully\n");
    return 0;
}
```

Output :-

```
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/File Manager$ gcc problem8.c -o problem8
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/File Manager$ ./problem8
File is Opened Successfully
How many Product Details You have to Enter : 5

Enter details of product 1
Product code: 101
Cost: 1
Number of items available: 50

Enter details of product 2
Product code: 102
Cost: 2
Number of items available: 10

Enter details of product 3
Product code: 103
Cost: 3
Number of items available: 40

Enter details of product 4
Product code: 104
Cost: 5
Number of items available: 10

Enter details of product 5
Product code: 105
Cost: 6
Number of items available: 40

Product details stored successfully
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/File Manager$
```

File Manager > product

	Code	Cost	Quantity
2	101	1.00	50
3	102	2.00	10
4	103	3.00	40
5	104	5.00	10
6	105	6.00	40
7			

9. Write a program to read the file created in problem8 and compute and print the total value of all the five products.

```
#include <stdio.h>

struct product
{
   int pcode;
   float cost;
   int quantity;
};

int main()
{
   FILE *fp;

   struct product p;
   float total = 0;
   char str[1000];

   fp = fopen("product", "r");
   if (fp == NULL)
   {
      printf("File is Not Open\n");
      return 1;
   }
   else
      printf("File is Opened Successfully\n");

   fgets(str, sizeof(str), fp); // Skip header line
   while (fscanf(fp, "%d %f %d", &p.pcode, &p.cost, &p.quantity) != EOF)
      total += (p.cost * p.quantity);

   fclose(fp);
   printf("Total value of all five products = %.2f\n", total);

   return 0;
}
```

Output :-

```
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/File Manager$ gcc problem9.c -o problem9
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/File Manager$ ./problem9
File is Opened Successfully
Total value of all five products = 480.00
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/File Manager$
```

10. Rewrite the program developed in problem8 to store the details in a random access file and print the details of alternate products from the file. Modify the program so that it can output the details of a product when its code is specified interactively.

```
#include <stdio.h>

struct product
{
   int product_code;
   float cost;
   int quantity;
} p;

int main()
{
   FILE *fp;

   int i, n, search_code, found = 0;

   fp = fopen("product.dat", "wb+");
   if (fp == NULL)
   {
      printf("File not opened\n");
      return 1;
   }

   /* Product Details */

   printf("How many Product Details You have to Enter : ");
   scanf("%d", &n);

   for (i = 0; i < n; i++)
   {
      printf("\nEnter product code: ");
      scanf("%d", &p.product_code);
      printf("Enter cost: ");
      scanf("%f", &p.cost);
      printf("Enter quantity: ");
      scanf("%d", &p.quantity);

      fwrite(&p, sizeof(p), 1, fp);
   }
```

```
    /* Alternate product details */

    printf("\nAlternate product details:\n");
    rewind(fp);

    for (i = 0; i < n; i += 2)
    {
        fseek(fp, i * sizeof(p), SEEK_SET);
        fread(&p, sizeof(p), 1, fp);

        printf("Code: %d  Cost: %.2f  Quantity: %d\n", p.product_code, p.cost, p.quantity);
    }

    /* search by product code */

    printf("\nEnter product code to search: ");
    scanf("%d", &search_code);

    rewind(fp);

    while (fread(&p, sizeof(p), 1, fp))
    {
        if (p.product_code == search_code)
        {
            printf("\nProduct Found:");
            printf("\nCode: %d", p.product_code);
            printf("\nCost: %.2f", p.cost);
            printf("\nQuantity: %d\n", p.quantity);
            found = 1;
            break;
        }
    }

    if (!found)
        printf("\nProduct not found\n");
    fclose(fp);

    return 0;
}
```

Output :-

```
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/File Manager$ gcc problem10.c -o problem10
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/File Manager$ ./problem10
How many Product Details You have to Enter : 5

Enter product code: 101
Enter cost: 2
Enter quantity: 20

Enter product code: 102
Enter cost: 5
Enter quantity: 10

Enter product code: 103
Enter cost: 10
Enter quantity: 15

Enter product code: 104
Enter cost: 15
Enter quantity: 30

Enter product code: 105
Enter cost: 20
Enter quantity: 21

Alternate product details:
Code: 101  Cost: 2.00  Quantity: 20
Code: 103  Cost: 10.00  Quantity: 15
Code: 105  Cost: 20.00  Quantity: 21

Enter product code to search: 102

Product Found:
Code: 102
Cost: 5.00
Quantity: 10
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/File Manager$
```

11. Write a C program that uses file handling methods to store records of mixed data in a file.

```
#include <stdio.h>

struct student
{
   int roll;
   char name[30];
   float marks;
}s;

int main()
{
   FILE *fp;

   int i, n;

   fp = fopen("records.txt", "w");

   if (fp == NULL)
   {
      printf("File is Not Open\n");
      return 1;
   }
   else
      printf("File is Opened Successfully\n");

   printf("Enter number of records: ");
   scanf("%d", &n);

   for (i = 0; i < n; i++)
   {
      printf("\nEnter details of student %d\n", i + 1);

      printf("Roll number: ");
      scanf("%d", &s.roll);
      getchar();
      printf("Name: ");
      scanf("%[^\n]s", s.name);
      printf("Marks: ");
      scanf("%f", &s.marks);
```

```
        fprintf(fp, "%d %s %.2f\n", s.roll, s.name, s.marks);
    }

    fclose(fp);
    printf("\nRecords stored successfully in file.\n");
    return 0;
}
```

Output :-

```
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/File Manager$ gcc problem11.c -o problem11
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/File Manager$ ./problem11
File is Opened Successfully
Enter number of records: 2

Enter details of student 1
Roll number: 1
Name: Ansh Kapadiya
Marks: 50

Enter details of student 2
Roll number: 2
Name: Anshul kamboj
Marks: 60

Records stored successfully in file.
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/File Manager$
```

12. Write a C program that uses getw function to read integer values from one file. Subsequently, it uses the putw function to write the integer values in reverse order in another file.

```
#include <stdio.h>

int main()
{
   FILE *fp1, *fp2;
   char f1[50], f2[50];
   int a[100], n = 0, i, value, num;

   printf("Enter name for Input File: ");
   scanf("%s", f1);
   fp1 = fopen(f1, "wb+");

   printf("How many integers do you want to enter? ");
   scanf("%d", &num);

   for (i = 0; i < num; i++)
   {
      printf("Enter integer %d: ", i + 1);
      scanf("%d", &value);
      putw(value, fp1); // Writing integer in binary
   }

   rewind(fp1);
   printf("\nEnter name for Output (Reverse) File: ");
   scanf("%s", f2);
   fp2 = fopen(f2, "wb+");

   if (fp1 == NULL || fp2 == NULL)
   {
      printf("File opening error!");
      return 1;
   }

   /* Read integers from file using getw() */
   while ((value = getw(fp1)) != EOF)
      a[n++] = value;

   /* Write integers in reverse order using putw() */
   for (i = n - 1; i >= 0; i--)
```

```
        putw(a[i], fp2);

    fclose(fp1);

    printf("\n--- Contents of %s (Reversed) ---\n", f2);
    rewind(fp2);
    while ((value = getw(fp2)) != EOF)
        printf("%d ", value); // Now printing to screen
    printf("\n");

    fclose(fp2);
    printf("Integers written in reverse order successfully in %s \n", f2);
    return 0;
}
```

Output :-

```
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/File Manager$ gcc problem12.c -o problem12
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/File Manager$ ./problem12
Enter name for Input File: input
How many integers do you want to enter? 2
Enter integer 1: 1
Enter integer 2: 5

Enter name for Output (Reverse) File: output.dat

--- Contents of output.dat (Reversed) ---
5 1
Integers written in reverse order successfully in output.dat
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/File Manager$
```

13. Write a C program that reads characters from a file and prints their ASCII codes.

```
#include <stdio.h>

int main()
{
   FILE *fp = fopen("input.txt", "r");

   if (fp == NULL)
   {
      printf("File cannot be opened\n");
      return 1;
   }

   printf("Character\tASCII Code\n");
   char ch;
   while ((ch = fgetc(fp)) != EOF)
      printf("%c\t\t%d\n", ch, (int)ch);

   fclose(fp);
   return 0;
}
```

Output :-

```
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/File Manager$ gcc problem13.c -o problem13
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/File Manager$ ./problem13
Character       ASCII Code
h               104
e               101
l               108
l               108
o               111
                32
W               87
o               111
r               114
l               108
d               100
                32
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/File Manager$
```

input.txt U

File Manager > input.txt

```
1   hello World
```

01

14. Write a C program that concatenates the contents of two files and writes then in the third file.

```
#include <stdio.h>

int main()
{
   FILE *fp1, *fp2, *fp3;

   char ch;

   fp1 = fopen("p14_1.txt", "r");
   fp2 = fopen("p14_2.txt", "r");

   if (fp1 == NULL )
   {
      printf("Cannot open p14_1.txt\n");
      return 1;
   }

   if (fp2 == NULL)
   {
      printf("Cannot open p14_2.txt\n");
      fclose(fp1);
      return 1;
   }

   fp3 = fopen("p14_3.txt", "w");
   if (fp3 == NULL)
   {
      printf("Cannot open p14_3.txt\n");
      fclose(fp1);
      fclose(fp2);
      return 1;
   }

   while ((ch = fgetc(fp1)) != EOF)
      fputc(ch, fp3);

   while ((ch = fgetc(fp2)) != EOF)
      fputc(ch, fp3);

   fclose(fp1);
```

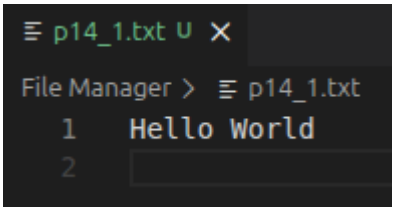
```
    fclose(fp2);
    fclose(fp3);

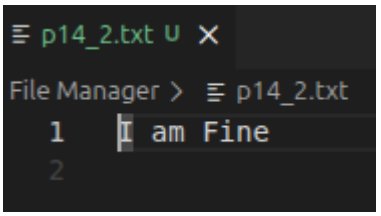
    printf("Files concatenated successfully.\n");

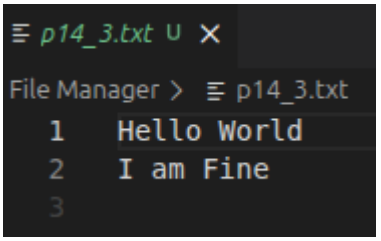
    return 0;
}
```

Output :-

```
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/File Manager$ gcc problem14.c -o problem14
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/File Manager$ ./problem14
Files concatenated successfully.
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/File Manager$
```







15. Write a C program that uses fscanf function to read integer values from a file, computes the square of each integer value and places the resultant values in a different file.

```
#include <stdio.h>

int main()
{
   FILE *fp1 = fopen("integer.txt", "r");
   if (fp1 == NULL)
   {
      printf("Cannot open integer.txt\n");
      return 1;
   }

   FILE *fp2 = fopen("square.txt", "w");
   if (fp2 == NULL)
   {
      printf("Cannot open square.txt\n");
      fclose(fp1);
      return 1;
   }

   int num;
   while (fscanf(fp1, "%d", &num) == 1)
      fprintf(fp2, "%d\n", num * num);

   fclose(fp1);
   fclose(fp2);

   printf("Successfully Written Squares in square.txt.\n");

   return 0;
}
```

Output :-

```
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/File Manager$ gcc problem15.c -o problem15
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/File Manager$ ./problem15
Successfully Written Squares in square.txt.
prince-kahar@prince-kahar:~/Prince/Sem 2 C Assignment/File Manager$
```

